# What produces a capturable complexity gap: one representative economy per model and group

Alexander Ober, with Claude

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**Scope.** Every economy with a protocol run (the committed record in `variants/results/economy_table.csv` plus the October 6–15, 2026 Quest campaign in `bop_runs/results`) and the reduced-form screens. The main tables hold one representative per model per group, built so that the same change is made in all three models; everything else that was run is in the appendix (Section 8). The tables are generated by `variants/campaign_tables.py` from the oracle and estimator files. Every campaign economy ran ten seeds (`gvypk3` twenty; the reduced form five per cell).

**Reading the tables.** Every table has the same columns. $\mathrm{SR}_{\max}$, the *linear ceiling* and the *room* are the population investor's conditional Sharpe ratios over the evaluation window: the linear ceiling is the best Sharpe a model linear in the ranks of the characteristics reaches with the true conditional moments, the room is the best nonlinear ceiling (random features, bins, quadratic) minus that linear ceiling, and *room % of linear* is the room as a percentage of the linear ceiling. That is the theoretical scope for nonlinearity, before any estimation. Sharpe ratios add in quadrature, so the room understates an interaction that sits on a large linear base; *interaction SR* is the nonlinear ceiling's Sharpe orthogonal to the linear one, $\sqrt{\mathrm{nl}^2 - \mathrm{lin}^2}$, the size of the interaction itself, and *DKKM* $\perp$ is DKKM's Sharpe orthogonal to the best classical method, signed like the gap. *Market share* is the equal-weighted market's share of $\mathrm{SR}_{\max}$. *DKKM*, *FMR* (Fama–MacBeth), *FF* (Fama–French factors), *linear* (the best of rank and level regressions, with and without the market) and *CAPM* (the estimated market line) are each method's out-of-sample Sharpe at the 360-month window. All Sharpe ratios are monthly (multiply by $\sqrt{12}$ for annual), so a gap of 0.01 is 0.035 a year. The *fair gap* is DKKM minus the best of the four classical methods, averaged across seeds, with in parentheses its $t$-statistic (the mean over its standard error across seeds; ten seeds unless stated); "+" counts the seeds in which it is positive. A dagger marks an oracle that predates the evaluation-window ceilings.

## 1 The answer

Each model has a spine: its published baseline, and one economy in which a priced state $y$ enters firm cash flows through an interaction of two firm variables (group 3b). Every other group is one change to that spine, the same change in all three models. Only the interaction produces a fair gap. With one type, every firm's exposure varies with one firm variable (3a), the premium is monotone in one characteristic's rank, and room is 5 to 10% of the linear ceiling with no gap; a crash-only factor (1) gives room of 3 to 12% and no gap; a quadratic price of the model's own risk (2) gives room of 0 to 14% and no gap; the interaction economies (3b) have room of 16 to 47% and gaps of +0.008 to +0.020 ($t$ 2.7 to 5.2); with the type loading's cross-sectional mean set to zero (`vyg25s`, `bvychs`,

Section 3b) room rises to 226% (KP14) and 20% (BGN) of the linear ceiling and the gap to +0.181 and +0.026, while GS21's mirrored ladder (`gvy25s`) gains room (2% to 11%) and no gap (−0.012). More priced factors (4) help only where each factor carries its own interaction: in the reduced form, factors that load on products of partly hidden states give room and a gap that rise with the number of factors and with the noise hiding the states, to +0.29 at ten factors, while factors that load linearly on observed characteristics give nothing at any $K$. The structural two-state economies (appendix) add no interaction: a second state through the same channel dilutes the exposure factor (KP14's room falls from 24% to 15%), and the designs that raise BGN's and GS21's gaps do so by giving each type its own state's premium, time variation in a sort rather than a second product.

The weight analysis (Section 6) shows the mechanism directly. In the 3b economies the true weight's slope in book-to-market depends on size (KP14, BGN) or on profitability (GS21); DKKM's fitted weight surface bends the same way, its nonlinear part correlating 0.77, 0.84 and 0.49 with the truth's, and in KP14 the interaction it recovers has the truth's size (0.107 against 0.106); Fama–MacBeth's surface is parallel lines by construction; and in the benchmarks there is no bend to find.

Economically: every factor model is firm exposure times aggregate shock, so the interaction that matters is not with the aggregate state but within the exposure. A capturable gap needs a firm's exposure to a priced shock to be the product of two things that different characteristics reveal, "what the firm is exposed to" (its type, or its productivity) times "how levered the firm's claim is to it" (growth options in KP14, the project portfolio in BGN, the coupon in GS21). What no linear portfolio holds is the part of the exposure orthogonal to every linear function of the characteristics, and with one priced factor that is all or nothing: a linear portfolio with any covariance with the exposure mimics the factor and takes its whole Sharpe (the reduced form's sign ladder, Section 3b). A type loading with a nonzero mean gives a sort that covariance, through the level compensation and the claim's leverage. That is why every positive-ladder structural gap is modest; why the GS21 economy in which leverage acts through the equity claim carries an interaction as large as KP14's (interaction SR 0.33) yet a near-zero gap, on a linear base of 0.81 (Section 3b); and why the imposed GS21 product (productivity times coupon, mean zero), the reduced form's latent products, and KP14's mirrored type ladder (`vyg25s`: loadings ±0.02 to ±0.14, mean zero, hidden by the level compensation) reach gaps of +0.02 to +0.3, +0.18 in vyg25s: the linear span has nothing of theirs to hold.

| group | the one change to the spine | room % of linear | fair gap |
|---|---|---|---|
| 1. distribution of shocks | a crash-only cash-flow factor | 3–12 | none |
| 2. price of risk | quadratic in the model's own state | 0–14 | none |
| 3a. single-characteristic exposure | one type exposed to $y$ | 5–10 | none |
| 3b. interaction | three types (a product in GS21) exposed to $y$ | 16–47 | +0.008 to +0.020 |
| 3b with a mean-zero type loading | the ladder mirrored, six types, level-compensated | 20 (BGN) / 226 (KP14) | +0.026 / +0.181 |
| the mean-zero loading doubled / tripled | `vyg25sx2`, `vyg25sx3` (KP14); `bvychsx2`, `bvychsx3` (BGN); `gvy25sx2` (GS21) | 343 / 175 (KP14); 26 / 21 (BGN); 13 (GS21) | +0.245 / +0.218; none; none |
| stronger BGN compensation | bv_comp 1.8 / 2.5 / 3.0 (`bvychs18`, `bvychs25`, `bvychs30`) | 27 / 67 / 67 | +0.006 / +0.048 / +0.063 |
| a non-monotone exposure | V-shaped loading across a visible level ladder (BGN `bvyvee`); quadratic in the firm's leverage (GS21 `gvyql10`, `gvyql20`, `gvyqh10`) | 23; 1–11 | none; none |
| a non-monotone exposure that works (3c) | quadratic in the firm's own productivity (GS21 `gvyzzh10`, `gvyzz20`); hump-shaped loading ±0.21 / ±0.28 across a mild size ladder (BGN `bvyhx3`, `bvyhx4`; KP14 `vyhx3`, `vyhx4`; GS21 `gvyhx3`) | 55–81 / 25–65; 37–40 / 59–77; 154–197 / 123; 9–31 | +0.080 / +0.059; +0.035 / +0.119; +0.212 / +0.166; +0.002 |
| a non-monotone exposure with a story (3c) | U-shaped loading across the size ladder (`bvyu4`, `vyu4`, `gvyu4`; reordered `bvyu5r`, `bvyu6r`, `vyu4r`, `vyu5r`); exposure U-shaped in leverage on fixed-coupon types (`gvylu30`, `gvylu30c`) | 54 / 28 / 2; 17 / 25 (BGN), 27 / 20 (KP14); 2 / 0 | +0.014 / +0.010 / stopped; +0.0295 / +0.025 (BGN), +0.003 / +0.000 (KP14); stopped |
| the interaction with a story in GS21 | industry sign × productivity (`gvysz1`, `gvysz2`); industry sign × fixed leverage (`gvysl15`, `gvysl20`) | 60 / 39; 1 | +0.053 / +0.059 ($t$ 11.5, ten of ten); stopped |
| the KP14 interaction slot | the positive ladder with loadings ×2 / ×3, price 3.5, bv_comp 2.5; the exposure on the profitability deviation (`vyge`, `vygel`, `vygesm`); the same exposure bounded (a capped logistic for $e^{by}$), mirrored types to ±1.2 on a size ladder (`vygelb` to `vygelbcs`) | 14–26 (vyg25: 19–28); 7–27; 26 / 36 / 51 / 56 / 49 / 69 / 80 / 90 | stopped at the oracle; stopped; +0.002 / +0.005 / +0.014 / +0.022 / +0.030 / +0.037 / +0.049 / +0.091 (`vygelbc`, $t$ 6.6; `vygelbcs`, $\sigma_\varepsilon$ 0.3, $t$ 12.3; both ten of ten) |
| 4. larger factor model (reduced form) | $K$ factors loading linearly on characteristics / on products of latent states | 0–1 / 190–660 | none / +0.03 to +0.29 |
| structural second priced state (appendix) | a second chain in each model | 15–65 | +0.005 to +0.056 |

The baselines every group is read against:

Table 1: Baselines. The three models as published, one type each: the floor every group is read against.

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bgnbase` – baseline, no priced state beyond the rate | 0.301 | 54% | 0.250 | +0.027 | 11 | 0.12 | 0.238 | 0.213 | 0.208 | 0.238 | 0.162 | -0.0001 (t -0.0) | +0.00 | 6/10 |
| GS21 | `gsbase` – baseline | 0.293 | 95% | 0.291 | +0.001 | 0 | 0.03 | 0.283 | 0.260 | 0.258 | 0.286 | 0.278 | -0.0028 (t -4.3) | -0.03 | 2/10 |
| KP14 | `kpbase` – baseline, one type, no priced state | 0.229 | 35% | 0.221 | +0.004 | 2 | 0.04 | 0.208 | 0.193 | 0.195 | 0.209 | 0.078 | -0.0010 (t -0.6) | -0.00 | 6/10 |

## 2 A different distribution of shocks: a crash-only factor

A cash-flow factor $y$ (unit variance, half-life two years) multiplies every firm's cash flows as $e^{by}$, with one exposure $b$ per model: 0.035 in BGN's project cash flows, $0.10\,\tilde{z}$ in GS21's output (the product economy's exposure without the coupon), 0.07 in KP14's project cash flows. Its Gaussian shocks carry no price. Its only priced event is a crash: each month, with probability 1%, $y$ jumps to $-3.5$ standard deviations, and the kernel weights that transition by $e^2$ (about 7% a month risk-neutral); $y$ then recovers at its ordinary mean reversion. Against the single-type economy of group 3a, the only change is that the Gaussian price is replaced by the crash.

Table 2: Group 1. A different distribution of shocks: a crash-only factor. Every firm's cash flows load on a factor y whose only priced shock is a rare crash (1% a month to $-3.5$ sd, kernel weight $e^2$).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvy1j` – crash-only factor: $e^{0.035y}$, the same crash | 0.324 | 50% | 0.278 | +0.033 | 12 | 0.14 | 0.265 | 0.239 | 0.235 | 0.262 | 0.160 | +0.0003 (t 0.1) | +0.01 | 6/10 |
| GS21 | `gvyz10j` – crash-only factor on $0.10\,\tilde{z}\,y$: y unpriced except the 1%/month jump to $-3.5$ (weight $e^2$) | 0.386 | 82% | 0.374 | +0.011 | 3 | 0.09 | 0.358 | 0.340 | 0.333 | 0.362 | 0.316 | -0.0046 (t -3.5) | -0.05 | 1/10 |
| KP14 | `vy1j` – crash-only factor: $e^{0.07y}$, y unpriced except a 1%/month jump to $-3.5$ (weight $e^2$) | 0.336 | 21% | 0.277 | +0.012 | 4 | 0.08 | 0.259 | 0.241 | 0.249 | 0.265 | 0.066 | -0.0057 (t -2.3) | -0.04 | 2/10 |

**Verdict.** A crash changes the distribution of a shock every firm bears in the same proportion. The cross-section of premia it creates is a sort on the one firm state that scales the exposure, so the linear ceiling takes nearly all of it: room is 12% of the linear ceiling in BGN, 3% in GS21 and 4% in KP14, and the fair gap is zero or negative in all three (+0.000, −0.005, −0.006). Adding the same crash to the interaction economies instead (Appendix A1) leaves their room and gap at the control's in BGN and GS21 and halves KP14's.

## 3 A different price of risk: quadratic in the model's own state

The price of each model's own aggregate risk is multiplied by $1 + 0.25\,s^2$, capped at 3, where $s$ is a standardized aggregate state: GS21's productivity $x$, and for KP14, whose own states are random walks, an unpriced stationary state $y$ that no firm is exposed to. The multiplier never falls below 1 because KP14's growth-option discount turns negative once the price of $x$-risk drops below about 0.82 of its value. In BGN the price of the market shock $\sigma_z$ is multiplied by the same function of the standardized rate, binned to seven equiprobable levels of the rate (1.00, 1.03, 1.17 and 1.68 from the middle outward) so that project values keep a closed form; the date-$k$ cash flow is priced at the bin of the rate one period before it pays.

Table 3: Group 2. A quadratic price of the model's own risk. The price of the baseline's aggregate risk becomes $1 + 0.25s^2$ times its value, capped at 3, $s$ a standardized aggregate state.

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bq25` – $\sigma_z \times \min(1 + 0.25\,s^2, 3)$, s the standardized rate, 7 bins | 0.411 | 36% | 0.346 | +0.049 | 15 | 0.18 | 0.339 | 0.303 | 0.293 | 0.335 | 0.149 | +0.0034 (t 0.7) | +0.00 | 4/10 |
| GS21 | `gq25` – $\gamma_x \times \min(1 + 0.25\,\tilde{x}^2, 3)$ | 0.345 | 95% | 0.343 | +0.002 | 0 | 0.03 | 0.334 | 0.316 | 0.309 | 0.337 | 0.326 | -0.0036 (t -5.5) | -0.05 | 0/10 |
| KP14 | `kpq` – $\gamma_x \times \min(1 + 0.25\,y^2, 3)$, y unpriced | 0.260 | 53% | 0.251 | +0.003 | 1 | 0.04 | 0.240 | 0.226 | 0.225 | 0.242 | 0.137 | -0.0021 (t -1.5) | -0.02 | 4/10 |

**Verdict.** A price of risk that moves with the aggregate state rotates the premium across states and leaves it linear in the ranks within one. The quadratic price gives room under 1% of the linear ceiling in GS21 and KP14, and 14% in BGN, where the rate is itself a priced state so a price that rises with its distance from the mean rotates the premium with it; the fair gap is zero in all three (+0.003, −0.004,

−0.002). The regime prices tried earlier (appendix) do the same, and BGN's fast regime gave the group's one small positive number (+0.008), which a ranks-times-state linear benchmark takes.

# 4 Different exposures to aggregate risk

## 4.1 Exposure that a single characteristic captures

The priced state $y$ is added with one type: every firm's cash flows load on it with the same exponent (the group 1 exposures, now with $y$'s Gaussian shocks priced at 2.5). A firm's exposure to $y$ then varies across firms only through one firm state: the mix of assets in place and growth option in BGN, productivity in GS21 (the product economy's exposure without the coupon), the growth-option share in KP14.

Table 4: Group 3a. Exposure captured by a single characteristic. A priced state y with one type: a firm's exposure varies only through one firm state (project mix, operating leverage, growth-option share).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvy1ch` – one type, project cash flows $\times e^{0.035y}$ | 0.336 | 54% | 0.297 | +0.029 | 10 | 0.13 | 0.283 | 0.252 | 0.255 | 0.281 | 0.181 | +0.0017 (t 0.8) | +0.02 | 7/10 |
| GS21 | `gvyz10` – one type, output $\times e^{0.10\,\tilde{z}\,y}$: the product economy without the coupon | 0.854 | 50% | 0.800 | +0.043 | 5 | 0.27 | 0.793 | 0.805 | 0.742 | 0.808 | 0.421 | -0.0164 (t -6.5) | -0.15 | 1/10 |
| KP14 | `vy1` – one type, cash flows $\times e^{0.07y}$ | 0.466 | 12% | 0.369 | +0.019 | 5 | 0.12 | 0.355 | 0.335 | 0.340 | 0.360 | 0.047 | -0.0050 (t -2.9) | -0.05 | 1/10 |

**Verdict.** A monotone function of one firm state, however nonlinear, is linear in the rank of the characteristic that reveals it. Room is 10% of the linear ceiling in BGN and 5% in GS21 and KP14, and the fair gap is zero or negative in all three (+0.002, −0.017, −0.005); in GS21 the productivity exposure alone carries a large premium ($SR_{max}$ 0.85) that the linear ceiling captures almost entirely. The same holds for every other single-characteristic wiring tried (Appendix A3), including ones with far more room: room is necessary, not sufficient. The absence of room is not an artefact of the rank standardization DKKM applies to its random features after the sines and cosines: recomputing the ceilings on the saved panels with the raw features (two seeds per economy, `variants/mlp/room_raw.py`) gives room of 4% / 6% / 8% of the linear ceiling in `vy1` / `gvyz10` / `bvy1ch` against 5% / 6% / 8% with the standardization, and 19% / 62% / 10% against 26% / 61% / 11% in the 3b economies; the quadratic and bin ceilings, which never standardize, say the same.

## 4.2 Exposure as an interaction of several characteristics

The same $y$ with heterogeneity in what is exposed to it: three types of firms with exposures 0.01 / 0.035 / 0.07 (BGN) and 0.02 / 0.07 / 0.14 (KP14), level-compensated so that value at $y = 0$ does not reveal the type; and in GS21, where types never produced room (Appendix A3), the product of standardized productivity and the standardized coupon written into the production function with coefficient 0.10. Exposure is then a product: type × project mix (BGN), type × growth-option share (KP14), productivity × leverage (GS21), each factor revealed by a different characteristic.

Table 5: Group 3b. Exposure as an interaction of characteristics. The same y with three types (BGN, KP14) or a product written into production (GS21): exposure = type or productivity × how levered the claim is.

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvych0` – y as a 21-node chain, three types (control) | 0.576 | 37% | 0.341 | +0.055 | 16 | 0.20 | 0.334 | 0.285 | 0.292 | 0.326 | 0.216 | +0.0076 (t 2.7) | +0.05 | 8/10 |
| GS21 | `gvyp10` – product exposure $\tilde{z}\tilde{b}$ at 0.10 (control) | 0.650 | 8% | 0.447 | +0.194 | 47 | 0.46 | 0.553 | 0.238 | 0.333 | 0.548 | 0.018 | +0.0057 (t 0.7) | +0.01 | 5/10 |
| KP14 | `vyg25` – three types on y, price 2.5 (control) | 0.495 | 11% | 0.346 | +0.082 | 24 | 0.25 | 0.350 | 0.292 | 0.320 | 0.335 | 0.044 | +0.0148 (t 5.2) | +0.09 | 10/10 |

**Verdict.** Every interaction economy has a positive mean gap with most seeds positive ($t$ of 2.7 to 5.1 across seeds), at room of 16 to 47% of the linear ceiling. The gaps are statistically clear and

economically modest: 2 to 4% of the linear Sharpe at these three representatives; the scaled GS21 versions in the appendix reach 11 to 22%. The premium is multiplicatively separable in the ranks, $f(A)\,g(B)$; a linear model in the ranks fits only additive functions, random features span products, and DKKM recovers 15 to 30% of the room as a gap. Going from 3a to 3b changes nothing but the second dimension of heterogeneity, and that is what creates the gap. The GS21 product ladder (Appendix A4) shows the room hump-shaped in the product's size and the gap rising to +0.08. The GS21 product is imposed: the coupon multiplies the exposure inside the production function. Its structural counterpart, where the exposure $0.10\,\tilde{z}\,y$ sits in cash flows and leverage acts only through the equity claim, was tested by fixing each firm's capital structure as a type drawn independently of productivity (`gvyzlf`: coupons 0.16, 0.42 and 0.68 per unit capital, debt-to-equity 0.25 to 3.25, no refinancing; four variants in the appendix). The interaction is there, the true premium's slope in leverage flips sign across profitability quintiles, but the equity claim amplifies the $y$ premium by only 20 to 40% around a large additive productivity part, so room stays at 6 to 8% of the linear ceiling and the gap at +0.005 ($t$ 1.8). The imposed product has no additive part, which is where its 43% comes from. Three levers on `gvyzlf` fail to change this (appendix). Valuing the debt as riskless (`gvyzlr`) leaves every number unchanged, so the debt was not absorbing $y$ risk; a more persistent $y$ (`gvyzlp`, $\kappa_y$ 0.10) halves $\mathrm{SR}_{\max}$ and the interaction with it; and cutting the price of the aggregate productivity shock from 0.5 to 0.1 (`gvyzlg`, with its own benchmark `gsbg1` and 3a comparator `gvyz10g`) lowers the market's share from 50% to 27% but leaves the linear ceiling at 0.72, because the base is the priced productivity level itself rather than the market: room 9%, gap −0.006. The interaction's own Sharpe is 0.31 to 0.33 in all of them, the size seen in KP14's and BGN's 3b economies (table column *interaction SR*); what differs is the base it sits on. Two further wirings close the question from the other side: fixing the product's sign with a leverage-level term (`gvyp10c`, exposure $0.10\,\tilde{b}(\tilde{z}+2)$) hands the market a large negative premium and leaves room of 1%, and KP14's own recipe, three latent exposure types times three fixed leverage types (`gvytl`, and `gvytlg` at a fifth of the aggregate shock's price), gives the market 72 to 84% of the attainable Sharpe and room of 1 to 3%. Every exposure with a nonzero cross-sectional mean is carried by GS21's equal-weighted market; the product is the one wiring whose mean is zero, which is why it alone has room.

**The sign of the type loading.** The dispersion bound in the summary is not what sets room, and the reduced form said so before the structural runs did. With one priced factor and $N = 500$, any linear-in-characteristics portfolio whose covariance with the exposure is not zero mimics the factor and diversifies the rest, so it takes the factor's whole Sharpe: room is all or nothing, and it is all only when the exposure is orthogonal to every linear function of the characteristics. The sign ladder of the reduced form (appendix, `lsh` rows) shows this directly: a product of two mean-zero latent states has room of +0.39 to +0.42, and shifting either factor's mean by one standard deviation, or more, or replacing the product by a one-signed ratio (`lrt`), sends room to 0.000 to 0.005 at the same $\mathrm{SR}_{\max}$. In the structural models the type loading's cross-sectional mean is that shift. Mirroring vyg25's ladder to six types of equal share with loadings ±0.02, ±0.07, ±0.14 (`vyg25s`, level-compensated as vyg25, price 2.5) raises KP14's room from +0.082 to +0.349, two and a quarter times a linear ceiling that falls from 0.346 to 0.174 (a mean-zero type leaves no $\bar{b}\times$growth-option piece for a book-to-market sort to hold), $\mathrm{SR}_{\max}$ from 0.495 to 0.575, and DKKM's fair gap from +0.015 to +0.181 ($t$ 17, ten of ten seeds; DKKM 0.373 against FMR's 0.102 and the linear model's 0.153). The market's share stays at 11%. BGN moves the same way but less far: `bvychs` (bvych0's ladder mirrored, ±0.01, ±0.035, ±0.07, level-compensated) has room of 20% of the linear ceiling against 16% and a fair gap of +0.026 ($t$ 4.9, nine of ten) against +0.008, with the linear ceiling unchanged (0.341 to 0.349) and $\mathrm{SR}_{\max}$ lower (0.576 to 0.510). Why BGN's linear ceiling holds where KP14's halves is not established; BGN's compensation is weaker ($\theta_f$ 1 to 1.60 against 1 to 3.36) and its $y$ is a bounded chain, so part of the type may stay visible in the characteristics. Hiding the type is what makes the sign matter: `vyg25u`, vyg25 without the level

compensation so that the type shows in value at $y = 0$, has vyg25's returns exactly ($SR_{max}$ 0.495 to every digit; only the characteristics change) and loses the gap ($-0.002$, $t$ $-0.5$, three of ten) and a fifth of the room, and `bvysg`, the earlier BGN signed ladder without compensation, has a negative gap where `bvychs` has a positive one. The capturable gap is then the Sharpe of the part of the exposure that no linear function of the characteristics spans, not a dispersion ratio, and a mean-zero type loading hidden by the level compensation puts nearly all of KP14's $y$ premium there. GS21 does not follow. `gvy25s`, gvy25's three level types mirrored to six of equal share with $b_f \in \{\pm 0.015, \pm 0.05, \pm 0.10\}$, level-compensated (bv_comp 1.2) at the same price, raises GS21's room from +0.014 to +0.062 (2% to 11% of the linear ceiling) and halves the market's share of $SR_{max}$ (84% to 50%), but the linear ceiling keeps 88% of $SR_{max}$ (0.558 of 0.632, against 0.684 of 0.739 in gvy25), and DKKM does not take what is left: the fair gap is $-0.012$ ($t$ $-4.4$, two of ten seeds positive; DKKM 0.549 against FMR's 0.541 and the fair linear model's 0.561), where the prediction was +0.02 to +0.06. The explanation this suggests, not yet tested, is the one thing that separates the three models: GS21's firm chooses its coupon, investment and default given its type, so a negative cash-flow loading shows in the chosen leverage and in book-to-market even though the level compensation hides it in value at $y = 0$, and the linear span holds most of the exposure; KP14's firm chooses nothing and BGN's only whether to invest, and the order of the three rooms (226%, 20%, 11%) is the order of how much the firm decides.

**The strength of the level compensation.** Two ladders test whether the compensation, which sets how visible the type is, moves room. In KP14 it changes the characteristics and nothing else: `vyg25s6` and `vyg25s18` (bv_comp 0.6 and 1.8) have `vyg25s`'s returns exactly, seed by seed. The type's visibility moves a lot ($R^2$ of $b_f$ on the ranks 0.67–0.74 at 0.6, through book-to-market; 0.88–0.92 at 1.2; 0.95–0.97 at 1.8, through size), the linear ceiling barely does (0.21, 0.17, 0.21), room is 147%, 226% and 167% of it, and the fair gap is +0.142 ($t$ 24), +0.181 and +0.185 ($t$ 8.7), five of five seeds at both new points. The premium stays hump-shaped in $b_f$ because the returns do not change, and the hump is the room; how visible the type is does not matter. In BGN the compensation does change returns, and weakening it goes the other way: `bvychs4`, `bvychs6`, `bvychs8` (bv_comp 0.4, 0.6, 0.8) make the type more visible through size ($R^2$ 0.89 at 0.4, 0.76 at 0.8) with the premium monotone in $b_f$, room falls from 20% to 12%, 8% and 7% of a linear ceiling that rises from 0.35 to 0.52 and 0.59, and the fair gap is $-0.004$, $+0.001$, $+0.009$, none distinguishable from zero (five, three and three seeds, the rest stopped at the oracle). The spec's prediction, a hidden sign near 0.6 lifting room above 50%, is falsified.

**GS21 with its capital structure fixed.** `gvytls` takes `gvytl`'s chassis (three coupons, 0.15 / 0.40 / 0.70 per unit capital, fixed for life and drawn independently of the type) and mirrors the exposure ladder as `gvy25s` does, eighteen mean-zero types, to test whether the firm's coupon choice is what reveals the sign in GS21. It is not: room is 8% of the linear ceiling (eight oracle seeds) against `gvy25s`'s 11%, and the fair gap $-0.007$ ($t$ $-1.0$, one of three; the other seven stopped). The type stays visible ($R^2$ 0.83–0.84; ROE +0.68, size +0.67, book-to-market $-0.63$), within each coupon the mean return rises with $b_f$, and the most exposed levered firms select themselves out: in each of three seeds none of the $b_f = -0.10$, coupon-0.70 firms and at most five of about thirty $b_f = -0.05$, coupon-0.70 firms survive into the panel. The prediction (room 15–25%) is falsified.

**A hump outside KP14, larger loadings, stronger compensation.** The decomposition of each type's premium on the market and on $y$ (`variants/hump_decomp.py`) says why only KP14's premium is hump-shaped: in `vyg25s` the market beta rises from 0.71 to 1.27 across the ladder while the residual premium falls from +0.10 to $-0.23$% a month, two opposed channels whose sum peaks inside; in `bvychs` both rise with $b_f$, in `gvy25s` they oppose but the $y$ channel is ten times the market channel. Seven economies tried to build the second channel into BGN and GS21 and none did: a pure curvature of GS21's productivity loading without $y$ (`gxq35`, `gxq70`) gives room of 1 to 3% and no gap; `gvy25s` with its level shifts kept and the $y$ price cut (`gvy25sq`, `gvy25sh`: $\gamma_v$ 0.25 and 0.5, bv_comp 12 and 9) makes the premium non-monotone but five times smaller, room 1 to 3%, gap $-0.004$ and $-0.008$.

BGN's compensation ladder continued in both directions (`bvychsn5`, `bvychs18`, `bvychs25`, `bvychs30`, `bvychs35`: bv_comp −0.5, 1.8, 2.5, 3.0, 3.5) never produces a hump, the premium staying monotone in $b_f$, but room rises from 20% of the linear ceiling at 1.2 and 27% at 1.8 to 67% at 2.5 and 3.0 and 66% at 3.5 (+0.16 to +0.19), cross-sectional nonlinearity in how the compensated characteristics map to $b_f$ (the bins basis alone reaches 0.36 against a linear 0.21). DKKM takes part of it: the fair gap is +0.048 ($t$ 3.7, nine of ten) at 2.5 and +0.063 ($t$ 5.6, ten of ten) at 3.0, the largest BGN gaps of the campaign, where 1.8 gives +0.006 and −0.5 gives −0.005 despite 35% room.

**The size of the loading.** The one lever that moved the gap is the size of the sign-switching loading, and only in KP14. `vyg25sx2` doubles `vyg25s`'s loadings (±0.04, ±0.14, ±0.28; $\theta_f$ 1 to 11.2): $\mathrm{SR}_{\max}$ 0.575 to 0.640, linear ceiling 0.174 to 0.158, room +0.437 (343% of the ceiling), fair gap +0.245 ($t$ 11.1, ten of ten; DKKM 0.456 against the linear model's 0.134) against +0.181; the hump deepens (seed 4: −0.04, +0.43, +0.60, +0.58, +0.43, +0.32% a month across the ladder, the residual channel +0.25 to −0.27 against a market beta of 0.58 to 1.36). Tripling (`vyg25sx3`; $\theta_f$ 1 to 37) is past the peak: the level spread begins to show linearly in size, the linear ceiling rises to 0.248, room is +0.364 (175%) and the gap +0.218 ($t$ 9.7). BGN and GS21 do not follow: `bvychsx2` and `bvychsx3` raise room from +0.068 to +0.089 and +0.081 with gaps of +0.007 and +0.002, and `gvy25sx2` makes the type linearly visible (linear ceiling 0.72 of an $\mathrm{SR}_{\max}$ of 0.82, gap +0.0005).

**Interaction and non-monotone exposures in every model.** The last round asked for six economies, in each model one whose gap comes from an interaction of characteristics and one whose exposure is non-monotone, each with a fair gap of 0.1 a year (+0.029 monthly). All six exist; the three non-monotone economies are in the next subsection. KP14 non-monotone: the hump across the size ladder, `vyhx3` (+0.212; the sign ladder's emergent hump `vyg25sx2`, +0.245, is its cousin here). BGN interaction: `bvychs35` (+0.071; the compensation ladder 2.5 / 3.0 / 3.5 / 4.0 gives +0.048 / +0.063 / +0.071 / +0.059, peaking at 3.5 because past it the level spread makes the type a size factor the Fama–French benchmark holds). GS21 interaction: the imposed product `gvyp15` (+0.078) and, with a story, the industry sign times the firm's productivity (`gvysz2`, +0.059, $t$ 11.5, ten of ten; `gvysz1` +0.053; the paragraph below); strong compensation on the mirrored ladder (`gvy25sc25`) gives room of 31–34% and no gap (−0.023). GS21 and BGN non-monotone: `gvyzzh10` (+0.080) and `bvyhx4` (+0.119), next subsection. KP14 interaction was the last filled, and ends as the largest interaction gap in any model: the sign-switching ladders' premium is hump-shaped whatever the compensation (`vyg25su`, `vyg25sux2`: bv_comp 0, the parents' returns to the digit, gaps +0.116 and +0.191), the positive ladder's levers leave room at `vyg25`'s 19–28% (loadings ×2 / ×3, $\gamma_v$ 3.5, bv_comp 2.5: 14–26%), and a new channel, the $y$ exposure on the firm's profitability deviation $b_f(\varepsilon - 1)$ (`bv_on_eps`; `vyge`, `vygel`, `vygesm`: the type hidden, revealed through size, or mirrored), leaves room at 7–27%: with positive loadings the product is a ROE sort with a type-dependent slope, and the mirrored version is capped at ±0.15 by the positivity of project values under the risk-neutral measure. The bounded exposure fills it: with $e^{b_f y}$ on the profitability deviation replaced by $h_b(y) = c/(1 + (c-1)e^{-cby/(c-1)})$ ($h(0) = 1$, slope $b$ at zero, range $(0, c)$; `bv_eps_cap`, the pricing equation unchanged but for $h$ on its right-hand side), the claim is bounded at $c$ times its unexposed value and the mirrored ladder runs at loadings to ±1.2 with positive project values. Room and gap rise together along the family (`vygelb` ±0.1/0.25/0.45, cap 2, price 2.5: room 21–31%, +0.002; `vygelb7` ±0.2/0.45/0.7: 27–45%, +0.005; `vygelb7g`, the same at price 4.0: 33–62%, +0.014; `vygelbw` ±0.3/0.7/1.2: 30–73%, +0.022; `vygelbx` ±0.5/1.2/2.0: 28–70%, +0.030, the width saturating), and `vygelbc` (±0.3/0.7/1.2 at cap 2.5, price 4.0) has room 40–90% and a fair gap of +0.037 ($t$ 6.6, ten of ten; DKKM 0.272 against 0.235): the type's sign read from size, the firm's deviation from ROE, their product spanned by no linear function of the characteristics. The profitability claim is a large share of value only for the negative loadings ($A_1/A_0$ 0.7–0.9 against 0.05–0.15), the room converts at about 17% of its size against 45% for the humps, and the levers are the cap (above 2.7 project values turn negative), the price of $y$ risk (`vygelbc6` at 6.0:

room 48–88%, +0.049, $t$ 5.8, ten of ten) and the ladder width, in that order; the strongest lever of all is the profitability dispersion, since the exposure $b_f(\varepsilon - 1)h'(y)$ scales with it: at $\sigma_\varepsilon$ 0.3 (`vygelbcs`, cap 2.5, price 4.0) the absolute room is 0.15–0.30 in Sharpe (58–119%) and the fair gap +0.091 ($t$ 12.3, ten of ten; DKKM 0.388 against 0.297; seeds +0.041 to +0.131), three times the bar. The reduced form agrees: a factor on the square of one partly hidden latent gives room +0.40 and a fair gap of +0.37 ($t$ 11) when the latent is book-to-market (`lsqb_a0_n0`), any shift of its mean adds a linear part that sends room to 0.002, and the same square on the size latent has the room and no fair gap because the value-weighted market inside the Fama–French set holds size[2].

**The interaction with a story in GS21.** The interaction economies share a frame, a priced shock with winners and losers by industry, the sign only partly revealed by size, times a firm-level amplifier, and differ in the amplifier: BGN's is the assets-in-place share (model-generated), KP14's the profitability deviation (a dispersion shock), and the imposed product `gvyp15` has no industry. GS21's version of the frame is the industry sign times the firm's standardized productivity, gs_bv$_f$ $\tilde{z}$ (gs_bzl 0, gs_bzi 1), six mirrored types on the new gs_level ladder 0.5–2.0, no compensation (`gvysz1`: ±0.1/0.2/0.35; `gvysz2`: ±0.15/0.3/0.5): a reallocation shock, the productive firms of a losing industry gaining share from the weak. gvysz1 has room 35–50% (0.22–0.30 in Sharpe) and a fair gap of +0.053 ($t$ 6.1, ten of ten; +0.023 to +0.107; DKKM 0.658 against 0.605), the type premium −0.30, +0.01, +0.11, +0.51, +0.42, +0.24% a month across the ladder, non-monotone in $b$; The larger ladder is stronger: gvysz2 has room 22–66% and a fair gap of +0.059 ($t$ 11.5, ten of ten; DKKM 0.678 against 0.619), GS21's interaction with a story. Financial leverage, the more natural amplifier, does not work as a type: two industries (gs_level 0.8 / 1.2) times three fixed coupons with the exposure proportional to the coupon (`gvysl15`, `gvysl20`) has room of 0.3–1% of a linear ceiling of 0.7–1.4 and was stopped; six perfectly observed groups are priced linearly.

### 4.3 Exposure that is non-monotone in a characteristic

A monotone exposure in one firm state is linear in that state's rank; an exposure that rises and then falls in a characteristic is not, and a linear model fits a line through the hump or the U. The construction, the same in every model that allows it: five equal types on a mild visible level ladder (the cash-flow or production level × 0.8, 0.9, 1.0, 1.1, 1.2; BGN's and KP14's type_level, GS21's new gs_level), each level-compensated at $y = 0$ (bv_comp 1.0), the loading hump-shaped, $b = -c, 0, +c, 0, -c$, or U-shaped, $+c$ at the ends, price 2.5. The U has the plain story (small firms exposed through financing constraints, large firms through operating leverage, mid-sized firms through neither); the hump's is thinner, the tails of the size distribution on one side of the shock and the middle on the other.

Table 6: Group 3c. Exposure non-monotone in a characteristic. The non-monotone constructions, by model: a hump-shaped loading ($b = -c, 0, +c, 0, -c$) or a U-shaped one ($+c$ at the ends) across five equal types on a mild visible level ladder (0.8, 0.9, 1.0, 1.1, 1.2 multiplies the cash-flow or production level), each type level-compensated at $y = 0$, price 2.5; GS21's exposure U-shaped in leverage on three fixed-coupon types; and GS21's exposure quadratic in the firm's own productivity. Rows marked with a dagger were stopped at the oracle.

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.551 | 28% | 0.349 | +0.155 | 44 | 0.36 | 0.375 | 0.295 | 0.280 | 0.340 | 0.153 | +0.0345 (t 3.9) | +0.14 | 10/10 |
| BGN | `bvyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.28, +0.00, +0.28, +0.00, -0.28 | 0.642 | 25% | 0.357 | +0.260 | 73 | 0.50 | 0.476 | 0.341 | 0.293 | 0.344 | 0.159 | +0.1186 (t 10.9) | +0.31 | 10/10 |
| KP14 | `vyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.632 | 7% | 0.207 | +0.377 | 197 | 0.54 | 0.431 | 0.217 | 0.136 | 0.178 | 0.031 | +0.2123 (t 12.5) | +0.37 | 10/10 |
| KP14 | `vyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.28, +0.00, +0.28, +0.00, -0.28 | 0.659 | 5% | 0.298 | +0.318 | 123 | 0.53 | 0.480 | 0.309 | 0.199 | 0.278 | 0.022 | +0.1664 (t 10.6) | +0.36 | 10/10 |
| GS21 | `gvyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2 (gs_level): $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.823 | 26% | 0.706 | +0.112 | 17 | 0.41 | 0.766 | 0.737 | 0.725 | 0.764 | 0.210 | +0.0023 (t 0.9) | +0.04 | 7/10 |
| GS21 | `gvyzzh10` – hump-shaped exposure in the firm's own productivity: $-0.1(\tilde{z}^2 - 1)$, one type | 0.639 | 25% | 0.364 | +0.250 | 70 | 0.49 | 0.480 | 0.325 | 0.255 | 0.400 | 0.152 | +0.0795 (t 4.6) | +0.25 | 10/10 |
| BGN | `bvyu4` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28, 0, 0, 0, +0.28 | 0.382 | 31% | 0.246 | +0.111 | 54 | 0.25 | 0.241 | 0.157 | 0.181 | 0.227 | 0.118 | +0.0142 (t 1.3) | +0.06 | 7/10 |
| BGN | `bvyu5r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.35 at the ends, unexposed reference type | 0.774 | 46% | 0.641 | +0.108 | 17 | 0.38 | 0.664 | 0.562 | 0.558 | 0.623 | 0.358 | +0.0295 (t 3.1) | +0.16 | 9/10 |
| BGN | `bvyu6r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.42 at the ends, unexposed reference type | 0.768 | 49% | 0.592 | +0.137 | 24 | 0.42 | 0.616 | 0.479 | 0.533 | 0.582 | 0.373 | +0.0246 (t 2.6) | +0.13 | 8/10 |
| KP14 | `vyu4` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28, 0, 0, 0, +0.28 | 0.371 | 13% | 0.263 | +0.072 | 28 | 0.20 | 0.256 | 0.216 | 0.209 | 0.246 | 0.040 | +0.0100 (t 1.5) | +0.04 | 7/10 |
| KP14 | `vyu4r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28 at the ends, unexposed reference type | 0.367 | 14% | 0.262 | +0.069 | 27 | 0.20 | 0.251 | 0.223 | 0.208 | 0.248 | 0.038 | +0.0025 (t 0.7) | +0.01 | 5/10 |
| KP14 | `vyu5r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.35 at the ends, unexposed reference type | 0.342 | 14% | 0.267 | +0.050 | 19 | 0.17 | 0.254 | 0.219 | 0.205 | 0.253 | 0.036 | +0.0003 (t 0.2) | -0.00 | 4/10 |
| GS21 | `gvyu4`† – U-shaped loading across a mild level ladder 0.8..1.2 (gs_level): $b$ = +0.28, 0, 0, 0, +0.28 | 0.917 | 85% | 0.893 | +0.020 | 2 | 0.18 | – | – | – | – | – | – | – | – |
| GS21 | `gvyl u30`† – exposure U-shaped in leverage: fixed coupons 0.15 / 0.40 / 0.70, $b$ = 0.3, 0, 0, 0.3, no compensation | 0.967 | 95% | 0.951 | +0.016 | 2 | 0.17 | – | – | – | – | – | – | – | – |
| GS21 | `gvyl u30c`† – exposure U-shaped in leverage: fixed coupons 0.15 / 0.40 / 0.70, $b$ = 0.3, 0, 0, 0.3, value-compensated | 1.219 | 99% | 1.213 | +0.005 | 0 | 0.11 | – | – | – | – | – | – | – | – |

**The hump across size: KP14 and BGN carry it, GS21 does not.** BGN's ladder rises with the

loading: room 9–13% at ±0.14, 37–40% and +0.035 ($t$ 3.9, ten of ten) at ±0.21, 59–77% and +0.119 ($t$ 10.9, ten of ten) at ±0.28 (`bvyhx2`, `bvyhx3`, `bvyhx4`). KP14's is the largest gap of this kind in the campaign: at ±0.21 (`vyhx3`) room 154–197% of a linear ceiling of 0.21–0.23 and a fair gap of +0.212 ($t$ 12.5, ten of ten; +0.134 to +0.306; DKKM 0.431 against 0.218), at ±0.28 (`vyhx4`) room 123% and +0.166 ($t$ 10.6, ten of ten). The type premium across the ladder is 0.36, 0.86, 0.68, 0.88, 0.35% a month (vyhx3, seed 0): the unexposed middle types earn the most, the market beta rises from 0.73 to 1.36 with $b$ while the $y$ channel pulls the other way, so the premium is non-monotone in the exposure itself and no characteristic that reveals the type can rank it. GS21's ladder gives nothing (`gvyhx3`: room 9–31%, +0.002, $t$ 0.9): its premium is monotone in the exposure (−0.50, +0.26, +1.47, +0.25, −0.56% a month), and a production level moves capital, value and profitability together, so book-to-market and ROE rank the exposure and the linear ceiling (0.4–0.9) holds it.

**The U in size.** At 0.28 the U is below the bar where the ladder works: `bvyu4` room 18–121%, +0.014 ($t$ 1.3, seven of ten); `vyu4` room 15–55%, +0.010 ($t$ 1.5, seven of ten); `gvyu4` room 1–5%, stopped. Two things hold it down. The compensation is referenced to the first type, here the small exposed one, whose positive exposure is discounted under $Q$, so every type was scaled to that low value; at 0.35 BGN's small type had no positive-NPV project and the gate rejected it (`bvyu5`); the reordered versions (`bvyu5r`, `bvyu6r`, `vyu4r`, `vyu5r`) put the unexposed level-1.0 type first. And, structurally, a one-signed exposure in a type ladder is revealed by the compensation itself: the compensated cash-flow level, hence ROE, ranks the types by $|b|$, and without compensation book-to-market does; the sign-flip hump escapes because its premium is non-monotone in the exposure. The reordered ladder shows it: in BGN the U reaches the bar at 0.35 (`bvyu5r`: room 8–27% of a linear ceiling that rose from 0.25 to 0.64, +0.0295, $t$ 3.1, nine of ten) and falls back at 0.42 (`bvyu6r`: +0.025, $t$ 2.6, eight of ten); in KP14 it has no gap at any loading (`vyu4r` +0.003, `vyu5r` +0.000), the compensation revealing the one-signed exposure completely.

**The U in leverage.** The most natural non-monotone story: highly levered firms in distress and lowly levered firms credit-rationed, both more exposed than the middle. On GS21's fixed-coupon chassis (`gvyzlf`: coupons 0.15 / 0.40 / 0.70 for life, D/E about 0.25 / 1.0 / 3.25, where the endogenous capital structure spans 0.002–0.006 across quintiles) a production-level exposure of 0.3 on the two outer types (`gvylu30`; `gvylu30c` value-compensated) has no room: 1–3% and 0.3–0.5% of linear ceilings of 0.5–1.4. With leverage fixed by type the exposure is a function of three perfectly observed groups, and a linear model in six characteristics fits three group means through leverage and book-to-market; room needs a second factor varying continuously within the type.

**The quadratic in productivity.** GS21's working non-monotone exposure: one type whose exposure to $y$ is quadratic in its own standardized productivity, $c\,(\tilde{z}^2 - 1)$, mean zero across firms (`gvyzzh10`, `gvyzz20`: $c = -0.1, +0.2$). The premium across ROE quintiles is a hump (−0.24, +0.58, +0.41, +0.34, −0.54% a month) or a U, room 55–81% and 25–65%, fair gap +0.080 ($t$ 4.6, ten of ten) and +0.059 ($t$ 6.3, ten of ten); the smaller U ($c = +0.1$) has the room and not the gap, the leverage characteristic revealing it. The story: firms of middling productivity are the most exposed, the most productive having the margin to absorb the shock, the least productive having already shed the capacity that bears it.

**Earlier attempts.** Before the ladder, putting the non-monotonicity into the exposure itself failed in both models. `bvyvee` gives BGN five types on a visible level ladder ($\theta_f = 0.6$ to 1.4 multiplying the cash-flow level) with the exposure V-shaped across it ($b_f = +0.07, 0, -0.07, 0, +0.07$): room 23%, gap +0.007 ($t$ 1.3), because the level itself moves returns monotonically (−0.15 to +1.0% a month along the ladder) and swamps the V. In GS21 a $y$ exposure quadratic in the firm's standardized coupon, $c(\tilde{b}^2 - 1)$ with $c = +0.1, +0.2, -0.1$ (`gvyql10`, `gvyql20`, `gvyqh10`), gives room of 1 to 11% and gaps of −0.002, −0.002, −0.026: the leverage characteristic spans only 0.002 to 0.006 across quintiles, so the

term has almost nothing to load on, and the one-type wiring hands the market a negative premium.

**The three economies and their stories.** KP14 `vyhx3` (+0.212) and BGN `bvyhx4` (+0.119), the hump across size, with BGN's U in size at the bar at 0.35 (`bvyu5r`, +0.0295); GS21 `gvyzzh10` (+0.080), the quadratic in productivity. In each, the exposure is a function of the characteristics that no linear combination of them spans, and the premium is non-monotone in whatever the characteristics reveal about the type; where the premium is monotone in a revealed quantity (the fixed-coupon leverage types, the GS21 size ladder, the one-signed U) the linear model holds it however curved the exposure is.

# 5 Larger factor models

More priced factors are tested where the structure is controlled: the reduced form below adds $K$ priced factors to a market factor, with the total price of risk held across $K$, and varies only whether the factors load linearly on the observed characteristics or on products of partly hidden states (setup in Section 7). The structural version, a second priced state in each of the three models, is in the appendix (Table 10).

Table 7: Reduced form: priced factors loading linearly on observed characteristics against products of latent states. Five seeds per cell, the total price of risk fixed across K (per-factor price $\propto 1/\sqrt{K}$); K = number of priced factors; the lfh rows load products of pairs of the five latent processes, observed with noise of sd 0 (n0) or 1 (n1).

| configuration | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kfh_lin_1 – K=1 factor on observed characteristics, linear loadings | 0.598 | 8 | 0.598 | +0.000 | 0 | 0.01 | 0.590 | 0.584 | 0.581 | 0.595 | 0.041 | -0.0067 (t -1.7) | 0/5 |
| kfh_lin_4 – K=4 | 0.592 | 8 | 0.588 | +0.003 | 0 | 0.06 | 0.555 | 0.578 | 0.567 | 0.575 | 0.041 | -0.0237 (t -3.1) | 0/5 |
| kfh_lin_16 – K=16 | 0.385 | 13 | 0.383 | +0.001 | 0 | 0.03 | 0.352 | 0.357 | 0.353 | 0.365 | 0.041 | -0.0128 (t -1.2) | 2/5 |
| lfh_L5_K1_n0 – K=1 product of latent states, no observation noise | 0.598 | 8 | 0.208 | +0.389 | 187 | 0.56 | 0.570 | 0.156 | 0.539 | 0.175 | 0.041 | +0.0311 (t 4.0) | 5/5 |
| lfh_L5_K2_n0 – K=2 | 0.598 | 8 | 0.144 | +0.453 | 314 | 0.58 | 0.564 | 0.111 | 0.520 | 0.115 | 0.041 | +0.0443 (t 6.7) | 5/5 |
| lfh_L5_K3_n0 – K=3 | 0.598 | 8 | 0.106 | +0.490 | 464 | 0.59 | 0.566 | 0.059 | 0.507 | 0.084 | 0.041 | +0.0589 (t 9.0) | 5/5 |
| lfh_L5_K5_n0 – K=5 | 0.598 | 8 | 0.092 | +0.502 | 549 | 0.59 | 0.543 | 0.048 | 0.423 | 0.064 | 0.041 | +0.1193 (t 21.6) | 5/5 |
| lfh_L5_K10_n0 – K=10 | 0.598 | 8 | 0.089 | +0.503 | 568 | 0.58 | 0.507 | 0.047 | 0.250 | 0.067 | 0.041 | +0.2572 (t 39.9) | 5/5 |
| lfh_L5_K1_n1 – K=1 product of latent states, observation noise 1.0 | 0.598 | 8 | 0.167 | +0.416 | 249 | 0.56 | 0.526 | 0.096 | 0.433 | 0.137 | 0.041 | +0.0930 (t 3.8) | 5/5 |
| lfh_L5_K2_n1 – K=2 | 0.598 | 8 | 0.111 | +0.469 | 423 | 0.57 | 0.523 | 0.072 | 0.420 | 0.082 | 0.041 | +0.1025 (t 6.8) | 5/5 |
| lfh_L5_K3_n1 – K=3 | 0.598 | 8 | 0.091 | +0.486 | 537 | 0.57 | 0.528 | 0.057 | 0.391 | 0.073 | 0.041 | +0.1371 (t 10.4) | 5/5 |
| lfh_L5_K5_n1 – K=5 | 0.598 | 8 | 0.075 | +0.496 | 664 | 0.57 | 0.496 | 0.038 | 0.309 | 0.060 | 0.041 | +0.1866 (t 26.3) | 5/5 |
| lfh_L5_K10_n1 – K=10 | 0.598 | 8 | 0.079 | +0.479 | 607 | 0.55 | 0.456 | 0.052 | 0.171 | 0.077 | 0.041 | +0.2854 (t 24.8) | 5/5 |

Factors that load linearly on observed characteristics leave no room at any $K$ and a negative gap; factors that load on products of partly hidden states give room and a gap that grow monotonically with $K$ and with the observation noise that hides the states. Bounded and curvature loadings on observed characteristics are in the appendix (the RF rows of Table 9).

# 6 What the estimated weights look like: the nonlinearity DKKM picks up

The Sharpe tables say DKKM beats the linear methods in the 3b economies; this section asks what its weights do that the linear methods' weights cannot. It applies the neural-network weight analysis (Pruitt's `MLP Analyze` pipeline, `external/MLP Analyze/BOP campaign/run_bop.py`) to the three benchmarks and the three 3b economies. For every economy, the per-firm weights of each method are rebuilt from the saved panels exactly as `run_estimators.py` formed them (`bop_runs/mlp/extract_weights.py`; the rebuild is accepted only if it reproduces the recorded per-month Sharpe ratios, and it does so to $10^{-16}$): the true SDF weights (the population investor's conditional mean-variance portfolio), the Fama–MacBeth weights, the Fama–French weights and the DKKM weights ($P = 3600$ random features, the two-draw ensemble, the economy's best $\kappa$), over the 125 evaluation months of every seed. A $5 \times 32$ ReLU network is then fitted to each method's weight as a function of the five characteristics, rank-standardized within each (seed, month) cross-section; the network is a smoother, and what is read

off it is the shape of the weight surface, not its fit. Three statistics summarize each surface against the truth's. The *correlation with truth* is the cross-sectional correlation of the method's weights with the true weights, averaged over cross-sections (first for the weights themselves, then for the fitted surface). The *nonlinear share* is the part of the fitted surface's variance that a linear regression on the five ranks within each cross-section leaves: the part of the surface no portfolio linear in the ranks can hold. The *correlation of nonlinear parts* is the cross-sectional correlation of that remainder with the truth's remainder: whether the method's departures from linearity are the truth's departures or noise. Its $t$-statistic is across seeds. Two figures per model show the surfaces themselves, in the two characteristics the economy's exposure is written in (size, the proxy for the firm's type, and book-to-market, its leverage, in KP14 and BGN; profitability and book-to-market in GS21), with the benchmark in the top row and the 3b economy in the bottom row and one panel per method. The *interaction* figure evaluates each fitted surface on a grid over the two characteristics (the other three at their mean), subtracts the best additive fit $f(a) + g(b)$, and maps what is left: the part of the weight that cannot be written as an effect of one characteristic plus an effect of the other. White is no interaction, a saddle (two opposite corners red, two blue) is a product term: the slope in book-to-market differs by size. Each panel has its own colour scale, so shapes are compared across panels and sizes read from the standard deviation in the panel title, which is also the table's last column. The *line* figure plots the fitted weight against one characteristic at five levels of the other; there an interaction shows as fanning or crossing lines, mixed with the additive part the interaction figure removes. Fama–MacBeth is linear in the characteristics, so its panels are parallel lines and its interaction is noise; Fama–French's portfolios are double sorts on size and each characteristic, so it carries an interaction by construction, but in the shape of its sort breakpoints.

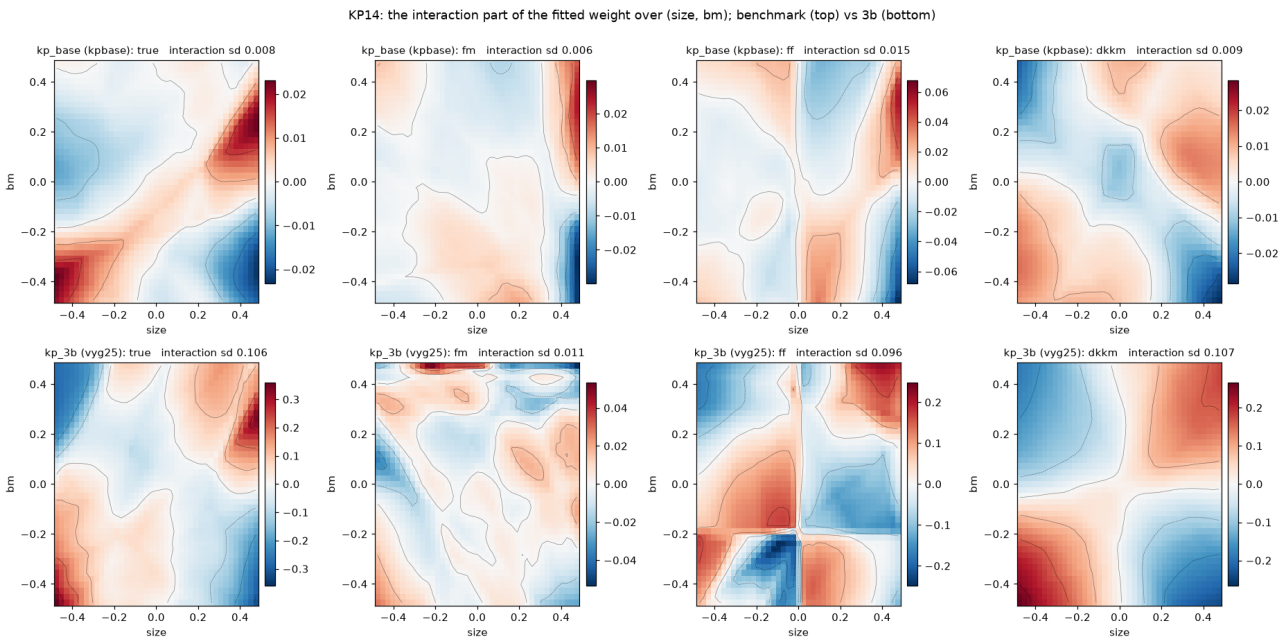



Figure 1: KP14: the interaction part of the fitted weight over book-to-market and size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

In every 3b economy the true weight is far from linear in the ranks (nonlinear share 0.39 to 0.58 of the surface), and the shape is an interaction: the weight's slope in one characteristic depends on the level of another. In KP14 (`vyg25`) the slope of the true weight in book-to-market depends on

| model | economy | weights | corr with truth | corr, MLP fit | nonlinear share | corr of nonlinear parts ($t$) | interaction sd at pair |
|---|---|---|---|---|---|---|---|
| BGN | `bgnbase` (benchmark; size × bm) | true SDF | 1.00 | 0.77 | 0.35 | | 0.076 |
| | | FMR | 0.43 | 0.57 | 0.11 | +0.28 (89.4) | 0.012 |
| | | FF | 0.35 | 0.44 | 0.35 | +0.11 (6.8) | 0.039 |
| | | DKKM | 0.45 | 0.60 | 0.04 | +0.37 (4.9) | 0.008 |
| BGN | `bvych0` (3b; size × bm) | true SDF | 1.00 | 0.38 | 0.39 | | 0.051 |
| | | FMR | 0.17 | 0.21 | 0.07 | +0.00 (0.4) | 0.013 |
| | | FF | 0.19 | 0.24 | 0.17 | +0.19 (14.2) | 0.019 |
| | | DKKM | 0.24 | 0.24 | 0.03 | +0.49 (59.9) | 0.013 |
| GS21 | `gsbase` (benchmark; bm × roe) | true SDF | 1.00 | 0.97 | 0.11 | | 0.004 |
| | | FMR | 0.54 | 0.74 | 0.07 | +0.13 (14.3) | 0.002 |
| | | FF | 0.48 | 0.69 | 0.39 | +0.30 (21.7) | 0.002 |
| | | DKKM | 0.57 | 0.79 | 0.14 | -0.12 (-8.3) | 0.000 |
| GS21 | `gvyp10` (3b; bm × roe) | true SDF | 1.00 | 0.98 | 0.50 | | 0.052 |
| | | FMR | 0.25 | 0.28 | 0.16 | +0.61 (68.8) | 0.021 |
| | | FF | 0.40 | 0.40 | 0.16 | +0.33 (17.8) | 0.025 |
| | | DKKM | 0.83 | 0.86 | 0.38 | +0.84 (83.9) | 0.033 |
| KP14 | `kpbase` (benchmark; size × bm) | true SDF | 1.00 | 0.80 | 0.08 | | 0.008 |
| | | FMR | 0.37 | 0.59 | 0.10 | +0.02 (1.4) | 0.006 |
| | | FF | 0.31 | 0.42 | 0.20 | +0.24 (17.3) | 0.016 |
| | | DKKM | 0.33 | 0.41 | 0.10 | -0.10 (-23.5) | 0.009 |
| KP14 | `vyg25` (3b; size × bm) | true SDF | 1.00 | 0.67 | 0.58 | | 0.112 |
| | | FMR | 0.29 | 0.33 | 0.19 | -0.13 (-18.2) | 0.012 |
| | | FF | 0.35 | 0.39 | 0.33 | +0.18 (16.6) | 0.095 |
| | | DKKM | 0.48 | 0.55 | 0.21 | +0.77 (229.0) | 0.110 |

Table 8: Neural-network fits of the portfolio weights: benchmark and group-3b economies.
Columns: corr with truth = mean over (seed, month) cross-sections of the correlation between the method's weights and the true SDF weights; corr, MLP fit = the same for the MLP surface fitted to the method's weights; nonlinear share = the part of the fitted surface's variance that an OLS fit on the five ranks within each cross-section leaves (one minus the within-cross-section $R^2$); corr of nonlinear parts = cross-sectional correlation of that nonlinear remainder with the truth surface's nonlinear remainder, with its t-statistic across seeds; interaction sd at pair = standard deviation, in weight units, of the two-way interaction part of the fitted surface over the named pair (the surface minus its additive fit in the two characteristics), so the truth's row says how large the interaction is and a method's row how much of it the method's weights carry. The pair is the one the economy's exposure is written in (KP14 and BGN: type, proxied by size, times leverage, bm; GS21: productivity, roe, times leverage, bm) and is the pair the figures use. Seeds: three per benchmark, ten per 3b economy; evaluation months only; DKKM at $P = 3600$ and the economy's best $\kappa$; weights winsorized at 1/99 within cross-sections.

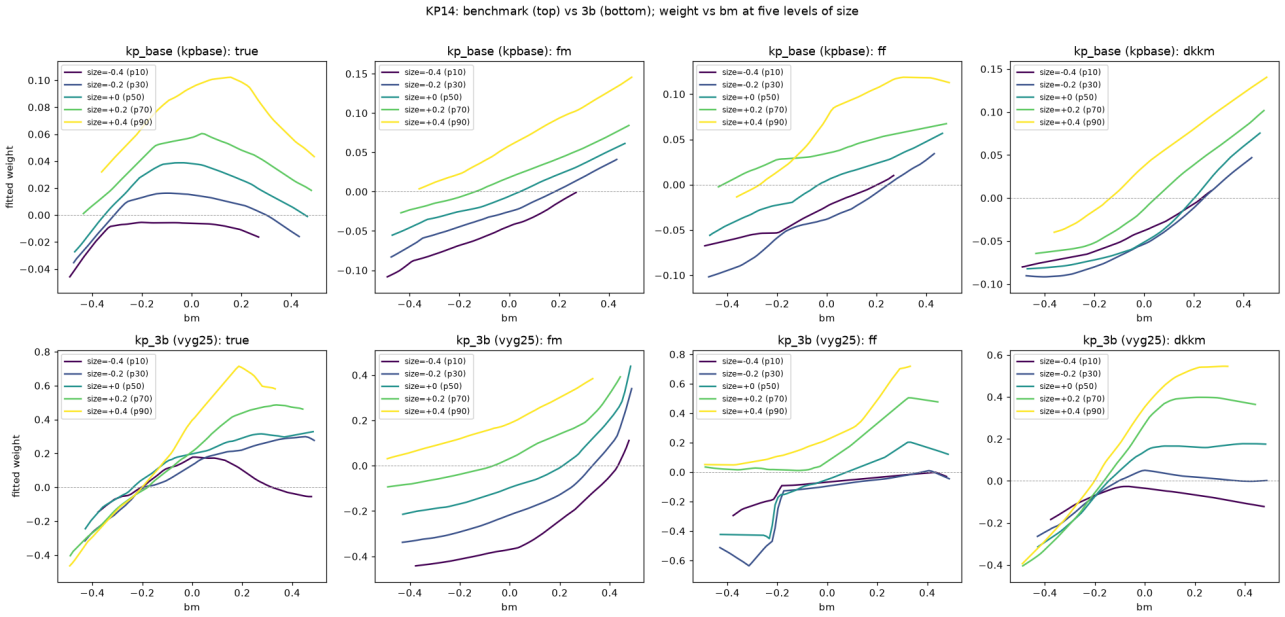



Figure 2: KP14: fitted weight against book-to-market at five levels of size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

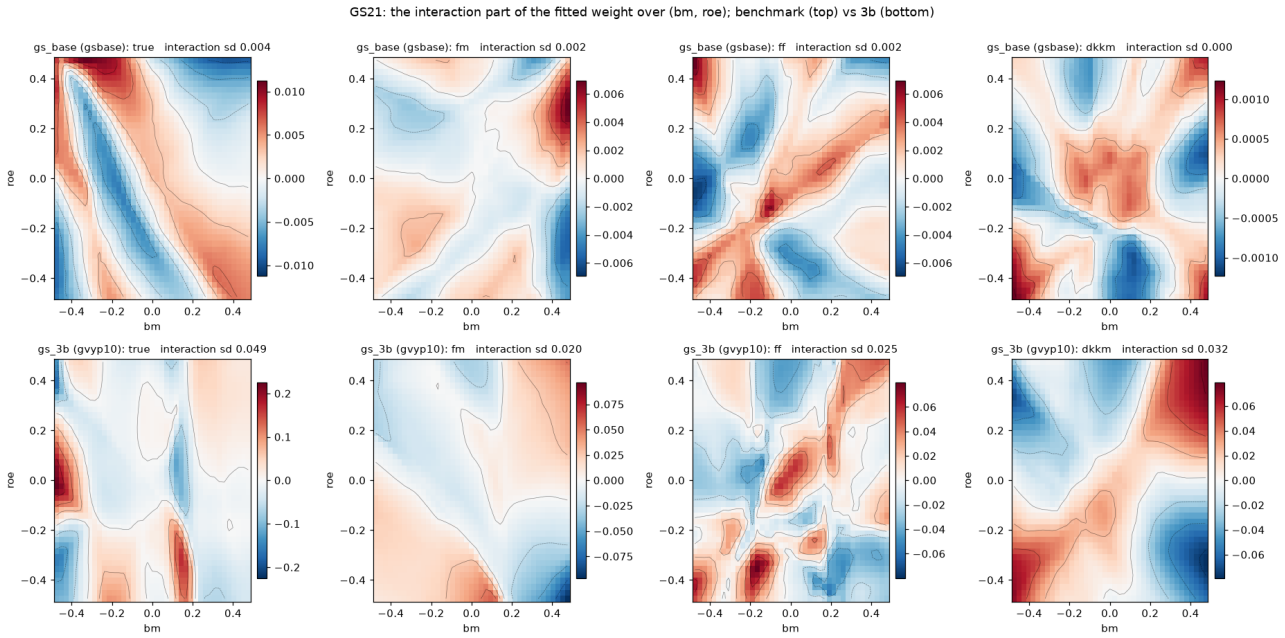



Figure 3: GS21: the interaction part of the fitted weight over profitability and book-to-market (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

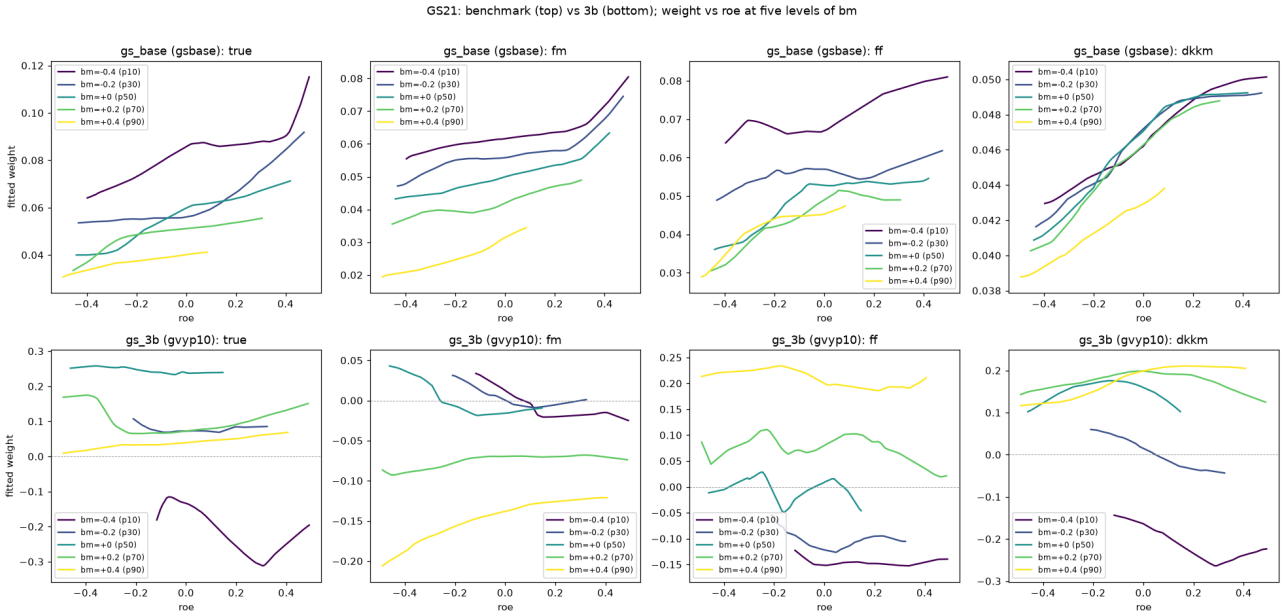



Figure 4: GS21: fitted weight against profitability at five levels of book-to-market (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

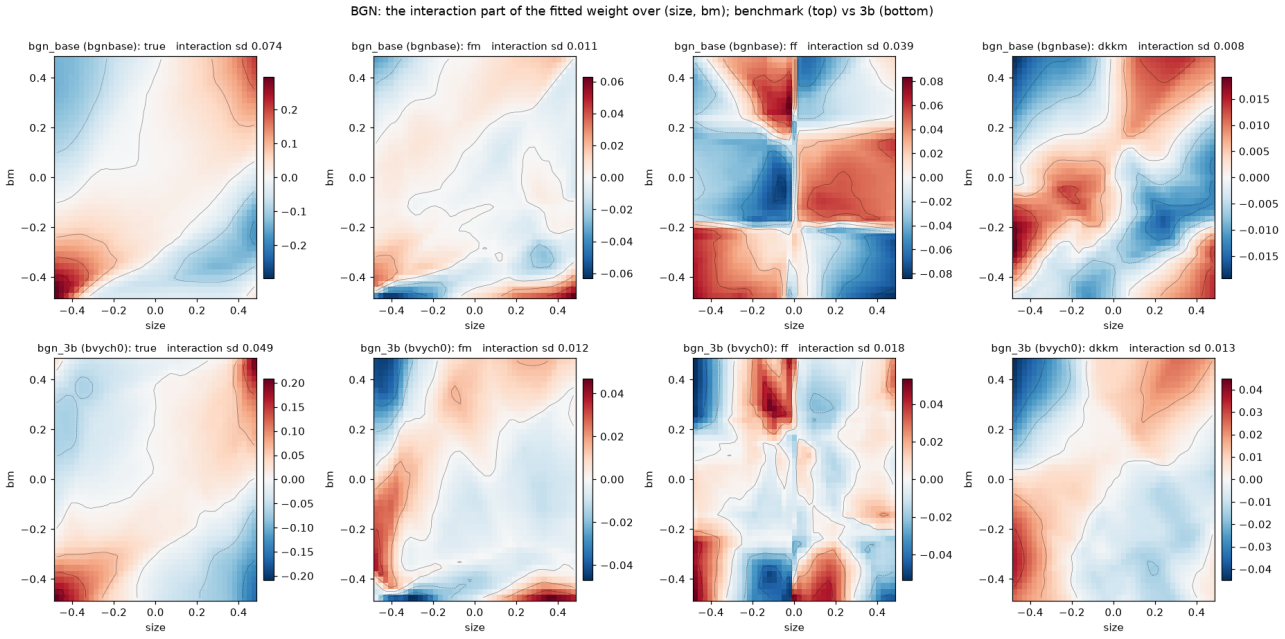



Figure 5: BGN: the interaction part of the fitted weight over book-to-market and size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

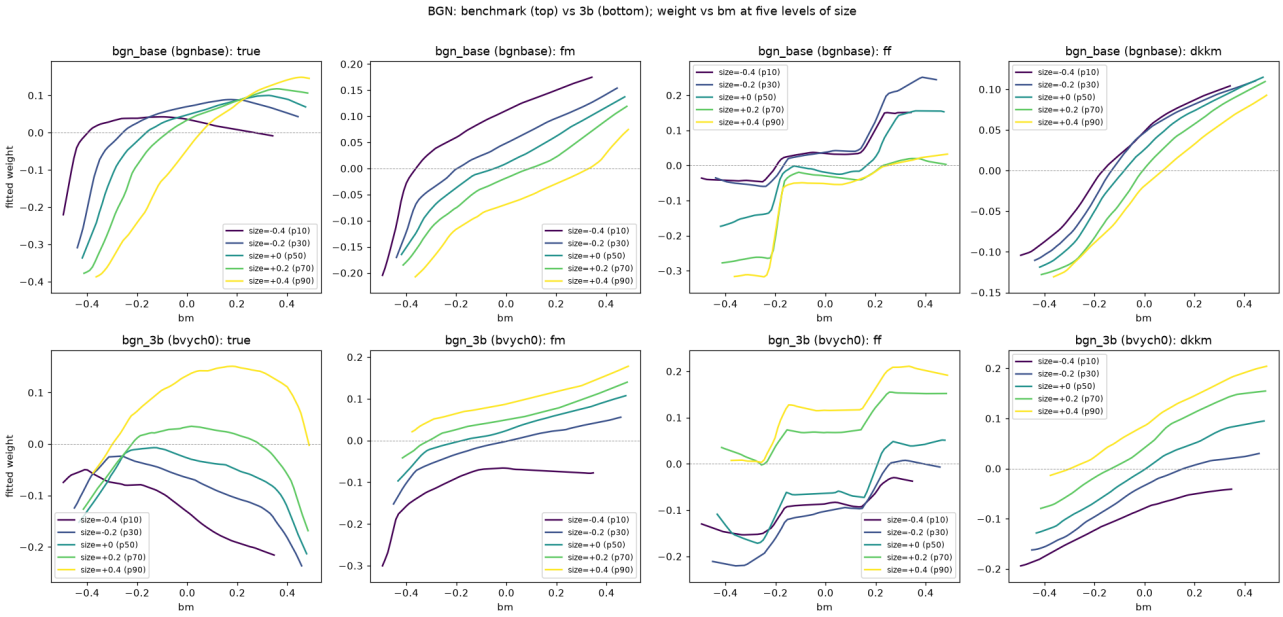



Figure 6: BGN: fitted weight against book-to-market at five levels of size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

the firm's size (the proxy for its type) and on its momentum: steep for large and high-momentum firms, flat for the rest. Over size and book-to-market the interaction part of the true surface has a standard deviation of 0.106 and DKKM's 0.107, with the same saddle (the interaction figure); DKKM's nonlinear part correlates 0.77 with the truth's. Fama–MacBeth's interaction is 0.011, parallel lines by construction, and its nonlinear part correlates −0.13. Fama–French's interaction is 0.096, because its portfolios are size-by-characteristic double sorts, but its shape is the sorts' steps rather than the truth's saddle and its nonlinear part correlates 0.18. The truth's larger interaction, book-to-market with momentum (sd 0.148), DKKM does not recover (0.015): momentum is the noisier proxy for the type. In GS21 (`gvyp10`) 98% of the true weight is a function of the five characteristics and the surface is non-monotone in book-to-market: median-bm firms get the largest weight, the lowest-bm firms a negative one with a V in profitability. The DKKM weights reproduce this ordering (correlation with truth 0.83 for the weights themselves, 0.84 between the nonlinear parts); Fama–MacBeth gets the ordering of the bm levels wrong (0.25), and Fama–French (0.40) has the steps without the curvature. Over book-to-market and profitability the interaction's sd is 0.049 in the truth and 0.032 in DKKM, with the same pattern, against 0.020 and 0.025 for Fama–MacBeth and Fama–French with different patterns. In BGN (`bvych0`) only 38% of the true weight is a function of the five characteristics, because the exposure sits on a latent firm type that the characteristics proxy weakly; what is a function of them is a hump in book-to-market that grows with size. The DKKM surface is nearly linear here (nonlinear share 0.03): the size-by-book-to-market interaction has sd 0.049 in the truth and 0.013 in DKKM, the right saddle at a quarter of its size, and DKKM's nonlinear part correlates 0.49 with the truth's where Fama–MacBeth's is uncorrelated (0.00) and Fama–French's 0.19. The departure from linearity it captures is small, which is the Sharpe table's smallest 3b gap. The benchmarks are the control. In KP14 (`kpbase`) and GS21 (`gsbase`) the true weight is close to linear in the ranks (nonlinear share 0.08 and 0.11): parallel humps in book-to-market at every size in KP14, parallel rises in profitability at every book-to-market level in GS21, with nothing for a nonlinear method to find. The interaction over the figure pair has sd 0.008 and 0.004 in the truth and 0.009 and 0.000 in DKKM, and DKKM's nonlinear part is uncorrelated with the truth's (−0.10 and −0.12): what it adds to a linear portfolio is noise,

which is the Sharpe tables' zero or negative gap, and in GS21 its best penalty is a hundred times the 3b economy's, shrinking it to the market. The BGN benchmark (`bgnbase`) is different: its true weight already has a size-by-book-to-market interaction (nonlinear share 0.35, interaction sd 0.074, the curves cross), and DKKM's surface is as linear there as in the 3b economy (nonlinear share 0.04, interaction sd 0.008); the BGN economies hand DKKM a nonlinearity it does not pick up in either version, which is why BGN's gaps are the smallest of the three models. Read across the figures, the mechanism is the same in all three models: DKKM's weights bend where the true weights bend, when the bend is an interaction of two characteristics large enough to show in the characteristics, and otherwise stay a linear portfolio plus noise.

# 7 What each economy changes

**Notation.** BGN (Berk, Green and Naik 1999): a firm is a portfolio of live projects; project $j$ has systematic risk $\beta_j$, drawn from an exponential distribution, and lognormal cash flows $C_j$; it survives each month with probability $\pi$; a new project arrives each month and is taken if its NPV is positive; the pricing kernel prices the market shock at $\sigma_z$ and the Vasicek short rate $r_t$. GS21 (Gomes and Schmid 2021): output is $e^{x_t+z_{it}}k_{it}$ with aggregate productivity $x_t$ and idiosyncratic productivity $z_{it}$ both AR(1); firms choose the coupon $b_{it}$, invest lumpily at a random cost, and may default; the kernel prices $x$-shocks at a constant $\gamma_x = 0.5$. KP14 (Kogan and Papanikolaou 2014): firm value is assets in place plus growth options; project cash flows are $x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, new projects arrive at the firm-specific rate $\lambda_{it}$, and the kernel prices disembodied productivity $x$ at $\gamma_x$ and embodied technology $z$ at $\gamma_z$.

**The added priced state $y$.** Every economy with a $y$ adds one stationary state with unit stationary variance and mean reversion $\kappa_y = 0.35$ per year, $y_{t+1} = \rho y_t + \sqrt{1-\rho^2}\,\eta_{t+1}$ with $\rho = e^{-\kappa_y/12}$ (an Ornstein–Uhlenbeck process in KP14, a Tauchen chain in GS21 and in BGN's chain economies). It is priced: the kernel is multiplied by $e^{-\lambda_y \eta_{t+1} - \lambda_y^2/2}$ with $\lambda_y = \gamma_v\sqrt{1/12}$, and $\gamma_v = 2.5$ unless stated. "Types" are equal shares of firms differing only in their exposure $b_f$ to $y$. "Level-compensated" means type $f$'s cash flows are also scaled by a constant $\theta_f = (V_0/V_f)^{1.2}$, the ratio of a unit project's value for type 0 and type $f$ at $y = 0$, so that value at $y = 0$ does not reveal the type (KP14's $\omega$ device).

- **Baselines**
  - `bgnbase`: BGN as above, one type. Expected returns come from the value-weighted $\beta_j$ of assets in place and from the growth option's exposure to $r_t$.
  - `gsbase`: GS21 as above, Table I calibration, one type.
  - `kpbase`: KP14 as above, one type, no $y$.
- **Group 1. A crash-only factor.** The 3a economy with the Gaussian price of $y$ set to zero ($\gamma_v = 0$) and a crash.
  - `bvy1j`, `gvyz10j`, `vy1j`: `bvy1ch`, `gvyz10` and `vy1` with $\gamma_v = 0$ and, each month with probability 1% whatever $y_t$, a jump of $y_{t+1}$ to $-3.5$ (the lowest node), weighted $e^2$ by the kernel (a risk-neutral probability of about 7% a month before renormalization). In KP14 the jump is a Poisson event at intensity 0.12 a year under $P$ and $0.12\,e^2$ under $Q$.
- **Group 2. A quadratic price of the model's own risk.**
  - `bq25`: `bgnbase` with $\sigma_z \to \sigma_z\, g(r_t)$, $g(r) = \min(1 + 0.25\,((r-\bar{r})/\mathrm{sd}_r)^2, 3)$ binned to seven equiprobable bins of the stationary rate ($g = 1.00$, 1.03, 1.17, 1.68 from the middle outward);

the date-$k$ cash flow is priced at the bin of $r_{t+k-1}$, through seven bases $D_j(r)$ in place of the regime's two.

- `gq25`: `gsbase` with $\gamma_x \to \gamma_x(1 + 0.25\,\tilde{x}^2)$ capped at 1.5, $\tilde{x} = x/\mathrm{sd}(x)$.
- `kpq`: `kpbase` with $\gamma_x \to \gamma_x \min(1 + 0.25\,y^2, 3)$, $y$ the unpriced stationary state of `kpgam` ($\kappa = 0.35$, unit variance); no firm is exposed to $y$.

- **Group 3a. One type exposed to $y$.**
  - `bvy1ch`: `bgnbase` plus $y$ (21-node chain), one type; project cash flows $C_j e^{0.035\,y_t}$.
  - `gvyz10`: `gsbase` plus $y$ (9-node chain over $\pm 3.5$ sd), one type; output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\,y_t} k_{it}$, $\tilde{z} = z/\mathrm{sd}(z)$: the product economy's exposure without the coupon.
  - `vy1`: `kpbase` plus $y$, one type; project cash flows $e^{0.07\,y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$.

- **Group 3b. Interactions.**
  - `bvych0`: `bvy1ch` with three types, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated ($\theta_f = (V_0/V_f)^{1.2}$). Exposure is $b_f$ times the firm's mix of assets in place and growth option.
  - `gvyp10`: `gsbase` plus $y$ (9-node chain), one type, output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$, with $\tilde{z} = z/\mathrm{sd}(z)$ and $\tilde{b} = (b - 0.5)/0.29$ the standardized coupon. Exposure is productivity $\times$ leverage, revealed by profitability and leverage.
  - `vyg25`: `vy1` with three types, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated. Growth options are options on the $e^{b_f y}$ cash flows, so a firm's exposure is $b_f$ times the elasticity of its value, which rises with its growth-option share: type $\times$ growth-option share, revealed by size and book-to-market.

- **Group 4. Two priced states.** Total price fixed, $\gamma_v = 2.5/\sqrt{2}$ on each.
  - `bvyk2`: `bvych0` with a second independent 21-node chain $y^{(2)}$; project cash flows

$$C_j\, e^{b_f y_t + b_f^{(2)} y_t^{(2)}}, \qquad (b_f, b_f^{(2)}) \in \{(0.01, 0),\ (0, 0.035),\ (0.049, 0.049)\}.$$

  - `gvypk3`: `gvyp10` with a second independent chain (5 nodes each, 25 joint) and three types of equal share, loading $0.10\,\tilde{z}\tilde{b}$ on $y$, on $y^{(2)}$, and $0.071\,\tilde{z}\tilde{b}$ on both.
  - `vy9t2`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; a firm's type is a pair $(f_1, f_2)$ drawn independently from $\{0.02, 0.07, 0.14\}$ for each state, nine types of equal share, and its cash flows load $e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each state.

## Reduced form (Table 7)

A simulated panel (`variants/reduced/rf_model.py`) of $N = 500$ firms over $T = 500$ months after a 100-month burn-in, five seeds per cell, run through the same oracle ceilings and the same estimators as the structural economies at the 360-month window; its true conditional moments are known in closed form. Notation: $i$ indexes firms, $t$ months, $j = 1, \ldots, 5$ the characteristics (size, book-to-market, asset growth, profitability, momentum), $k = 1, \ldots, K$ the extra priced factors.

- **Latent processes.** Each firm carries five independent AR(1) processes $z_{ijt} = \rho_j z_{ij,t-1} + \sqrt{1 - \rho_j^2}\,\eta_{ijt}$, $\eta$ standard normal, $\rho = (0.99, 0.98, 0.90, 0.95, 0.70)$, so each has unit stationary variance.

- **Observed characteristics.** $X_{ijt} = z_{ijt}$ in the `kfh` and `kfo` ladders. In the `lfh` ladder the five processes are the firm's true states and each characteristic is its state plus observation noise, $X_{ijt} = z_{ijt} + \sigma_\nu \nu_{ijt}$, with $\nu_{ijt}$ an independent AR(1) with the same $\rho_j$ and unit variance. The ladder's one noise parameter is $\sigma_\nu$: 0 (the states are observed exactly, rows `n0`) or 1 (noise as large as the state itself, rows `n1`). $\tilde{X}_{it}$ is $X_{it}$ standardized across firms at $t$. The oracle's bases and the estimators see the ranks of $X_{it}$, as everywhere in the report.

- **Returns.** $r_{i,t+1} = B_{it}\lambda + B_{it} f_{t+1} + \varepsilon_{i,t+1}$, with $f_{t+1} \sim N(0, \sigma_f^2 I_{K+1})$, $\sigma_f = 0.04$, and $\varepsilon_{i,t+1} \sim N(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon = 0.10$, all independent. $B_{it}$ is the $1 \times (K+1)$ row of firm $i$'s loadings and $\lambda$ the $(K+1)$-vector of constant prices of risk. The true premium is $\mu_{it} = B_{it}\lambda$ and the true covariance $\Sigma_t = \sigma_f^2 B_t B_t' + \sigma_\varepsilon^2 I$.

- **The market factor (column 0).** $B_{i0t} = 1 + a'\tilde{X}_{it}$ with $a = (0, 0.3, 0, 0.3, 0.3)$ on (size, bm, agr, roe, mom), and $\lambda_0 = 0.002$, a Sharpe of $\lambda_0/\sigma_f = 0.05$. This loading is linear in the characteristics, so every method holds it.

- **The $K$ extra factors (columns $k = 1, \ldots, K$).** $\lambda_k = 0.024/\sqrt{K}$ for every $k$, so each has Sharpe $0.6/\sqrt{K}$ and the total price of risk is the same at every $K$. Every extra loading is centered across firms each month ($\bar{\beta}_{kt}$ is the cross-sectional mean), so the equal-weighted market carries none of the extra premia. The ladders differ only in the loading:

  - ***`kfh`*** (loadings on observed characteristics): $B_{ikt} = h(w_k'\tilde{X}_{it}) - \bar{\beta}_{kt}$, with $w_k$ a fixed unit vector in $\mathbb{R}^5$ drawn once and reused in every seed, and $h(u) = u$ in the main table (lin); the appendix adds $h(u) = \tanh(u)$ (tanh) and $h(u) = u^2$ (sq). Exposure is a function of what the characteristics show directly: GS21's structure.
  - ***`kfo`*** (appendix): `kfh` with the size coordinate of each $w_k$ set to zero before normalizing, so no extra factor loads on size.
  - ***`lfh`*** (loadings on products of latent states): $B_{ikt} = z_{ia_kt}\, z_{ib_kt} - \bar{\beta}_{kt}$, where $(a_k, b_k)$ is the $k$-th pair in a fixed enumeration of the ten pairs of the five processes (so $K \leq 10$). The loading is a product of two true states that the characteristics reveal only up to the observation noise $\sigma_\nu$: KP14's and BGN's structure, a type times a within-firm state.

The ladders therefore vary one thing at a time: $K$; whether the loading is a function of the observed characteristics or a product of partly hidden states; and, within `lfh`, how much noise hides the states. In the `kfh` linear and tanh ladders $\mathrm{SR}_{\max}$ falls at $K = 16$ because sixteen directions over five characteristics are collinear and their loadings partly cancel; in the `lfh` ladder $\mathrm{SR}_{\max}$ stays near 0.6 at every $K$.

# 8 Additional economies

Everything else that was run. The table has the main tables' columns; the bullets describe each economy against its parent (notation as in Section 7). Three groups of them answer side questions: the crash added to the interaction economies instead of to a single-type economy (`bvyj25`, `gvyp10j`, `vyj25`), the GS21 product ladder (`gvyp05` to `gvyp20`, `gvyp10x`), and the earlier single-characteristic and price-of-risk wirings that the harmonized cells replace.

**BGN**

- `g0235f`: `bgnbase` with $\sigma_z \to g_s \sigma_z$, $s_t \in \{\text{calm}, \text{stress}\}$ a Markov chain, $g = 0.2$ calm and 3.5 stress; monthly switch probabilities 1/12 (calm to stress) and 1/6 (stress to calm), so stress is one third of months with a mean spell of 6 months.

Table 9: Additional economies. Everything else that was run, by model, each described in the appendix against its parent; the RF rows are the reduced-form ladders with bounded (tanh), curvature (square) and size-orthogonal curvature (kfo) loadings, the sign ladder (lsh: the K=1 latent product with the factors' means shifted; lrt: a one-signed ratio), and the curvature-in-a-latent ladder (lsq / lsqb: the square of one partly hidden latent, size or book-to-market, with its mean shifted).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `g0235f` – regime price ×0.2 / ×3.5, spells 4× faster | 0.237 | 62% | 0.163 | +0.045 | 28 | 0.13 | 0.169 | 0.141 | 0.131 | 0.161 | 0.146 | +0.0079 (t 4.8) | +0.05 | 10/10 |
| BGN | `g0235df` – stress 2/3 of the time, 4× faster | 0.489 | 17% | 0.432 | +0.036 | 9 | 0.18 | 0.423 | 0.392 | 0.351 | 0.428 | 0.077 | -0.0051 (t -2.2) | -0.05 | 2/10 |
| BGN | `bryc25` – three types on y, level-compensated (1.2) | 0.601 | 41% | 0.398 | +0.062 | 16 | 0.23 | 0.399 | 0.344 | 0.367 | 0.384 | 0.248 | +0.0148 (t 3.0) | +0.09 | 9/10 |
| BGN | `brych61` – brych0 on a 41-node chain | 0.565 | 38% | 0.336 | +0.052 | 15 | 0.19 | 0.328 | 0.279 | 0.291 | 0.320 | 0.215 | +0.0086 (t 2.8) | +0.06 | 8/10 |
| BGN | `bryc0` – the same without compensation | 0.475 | 41% | 0.395 | +0.066 | 18 | 0.22 | 0.386 | 0.321 | 0.347 | 0.376 | 0.195 | +0.0104 (t 3.1) | +0.07 | 9/10 |
| BGN | `bryj25` – the chain with a priced crash (1%/month to $-y_{max}$, priced $e^2$) | 0.536 | 38% | 0.315 | +0.054 | 17 | 0.19 | 0.307 | 0.261 | 0.269 | 0.301 | 0.202 | +0.0066 (t 2.8) | +0.05 | 9/10 |
| BGN | `brysg` – signed exposure ladder ($-0.07$ / 0 / $+0.07$) | 0.504 | 28% | 0.361 | +0.120 | 35 | 0.31 | 0.318 | 0.305 | 0.273 | 0.324 | 0.141 | -0.0199 (t -2.0) | -0.08 | 2/10 |
| GS21 | `g2B` – countercyclical $\gamma(x)$, clipped | 0.314 | 95% | 0.312 | +0.001 | 0 | 0.02 | 0.303 | 0.254 | 0.256 | 0.305 | 0.298 | -0.0024 (t -4.3) | -0.04 | 9/10 |
| GS21 | `bx7` – two-state regime on $\gamma$, five types on x | 0.327 | 94% | 0.317 | +0.007 | 2 | 0.07 | 0.309 | 0.282 | 0.271 | 0.311 | 0.306 | -0.0017 (t -2.8) | -0.03 | 0/10 |
| GS21 | `gx7` – five types on productivity x | 0.485 | 98% | 0.474 | +0.008 | 2 | 0.08 | 0.473 | 0.393 | 0.383 | 0.473 | 0.473 | +0.0003 (t 1.6) | +0.01 | 5/10 |
| GS21 | `gxy1` – one type, output $\propto e^{0.05y}$ | 0.636 | 95% | 0.633 | +0.002 | 0 | 0.05 | 0.626 | 0.608 | 0.598 | 0.629 | 0.602 | -0.0030 (t -4.0) | -0.05 | 1/10 |
| GS21 | `gxy1j` – crash-only factor: $e^{0.05y}$, the same crash | 0.379 | 96% | 0.377 | +0.001 | 0 | 0.03 | 0.372 | 0.350 | 0.341 | 0.374 | 0.363 | -0.0022 (t -4.4) | -0.04 | 1/10 |
| GS21 | `gxy25` – production-level exposure to y, three types | 0.725 | 86% | 0.709 | +0.014 | 2 | 0.14 | 0.688 | 0.682 | 0.666 | 0.695 | 0.625 | -0.0070 (t -4.0) | -0.09 | 1/10 |
| GS21 | `gxyz25` – signed fixed-cost exposure to y | 0.619 | 69% | 0.602 | +0.014 | 2 | 0.13 | 0.496 | 0.490 | 0.426 | 0.499 | 0.420 | -0.0032 (t -1.5) | -0.03 | 2/6 |
| GS21 | `gxyp05` – product $z\tilde{z}$ b 0.05 | 0.425 | 60% | 0.284 | +0.128 | 45 | 0.30 | 0.297 | 0.254 | 0.202 | 0.276 | 0.254 | +0.0203 (t 7.4) | +0.11 | 10/10 |
| GS21 | `gxyp075` – product at 0.075 | 0.561 | 28% | 0.261 | +0.287 | 111 | 0.48 | 0.433 | 0.212 | 0.187 | 0.352 | 0.149 | +0.0802 (t 9.4) | +0.25 | 10/10 |
| GS21 | `gxyp15` – product at 0.15 | 0.744 | -25% | 0.616 | +0.119 | 22 | 0.39 | 0.665 | 0.349 | 0.516 | 0.612 | 0.192 | +0.0528 (t 9.3) | +0.26 | 10/10 |
| GS21 | `gxyp20` – product at 0.20 | 0.778 | -51% | 0.729 | +0.039 | 6 | 0.23 | 0.710 | 0.452 | 0.643 | 0.715 | 0.407 | -0.0049 (t -1.1) | -0.04 | 4/10 |
| GS21 | `gxyp10x` – product at 0.10 plus a curvature loading on x | 0.589 | 17% | 0.365 | +0.215 | 63 | 0.45 | 0.491 | 0.196 | 0.262 | 0.448 | 0.054 | +0.0431 (t 7.6) | +0.20 | 10/10 |
| GS21 | `gxyp10j` – the product exposure with a priced crash | 0.537 | 21% | 0.328 | +0.197 | 62 | 0.41 | 0.442 | 0.242 | 0.246 | 0.415 | 0.072 | +0.0267 (t 3.8) | +0.14 | 10/10 |
| GS21 | `gxyp2` – two priced chains, two product types | 0.478 | 27% | 0.226 | +0.212 | 103 | 0.37 | 0.309 | 0.168 | 0.153 | 0.254 | 0.105 | +0.0559 (t 3.2) | +0.15 | 9/10 |
| GS21 | `gxyz1` – productivity exposure, three fixed-coupon leverage types (diluting) | 0.865 | 50% | 0.793 | +0.059 | 8 | 0.31 | 0.804 | 0.793 | 0.743 | 0.799 | 0.426 | +0.0049 (t 2.5) | +0.05 | 7/10 |
| GS21 | `gxyz1f` – productivity exposure, three leverage types, capital structure fixed for life | 0.880 | 50% | 0.808 | +0.066 | 8 | 0.33 | 0.820 | 0.807 | 0.758 | 0.815 | 0.432 | +0.0046 (t 2.0) | +0.05 | 7/10 |
| GS21 | `gxyz1v` – gyzylf with coupons 0.05 / 0.45 / 0.85 | 0.782 | 52% | 0.729 | +0.048 | 7 | 0.27 | 0.722 | 0.730 | 0.676 | 0.738 | 0.404 | -0.0165 (t -5.5) | -0.15 | 0/10 |
| GS21 | `gxyz1fv` – gyzylf with five leverage types | 0.880 | 50% | 0.812 | +0.060 | 7 | 0.32 | 0.818 | 0.812 | 0.765 | 0.818 | 0.440 | -0.0016 (t -0.7) | -0.03 | 4/10 |
| GS21 | `gxyz1b` – gyzylf with the exposure doubled | 1.066 | 45% | 1.003 | +0.056 | 7 | 0.31 | 0.996 | 0.968 | 0.944 | 0.985 | 0.490 | +0.0114 (t 1.5) | +0.06 | 6/10 |
| GS21 | `gxyz1r` – gyzylf with riskless debt | 0.880 | 49% | 0.808 | +0.066 | 8 | 0.33 | 0.820 | 0.807 | 0.758 | 0.815 | 0.432 | +0.0045 (t 2.0) | +0.05 | 7/10 |
| GS21 | `gxyz1p` – gyzylf with a persistent $y$ ($\kappa_y$ 0.10) | 0.475 | 76% | 0.457 | +0.017 | 4 | 0.13 | 0.448 | 0.428 | 0.418 | 0.451 | 0.355 | -0.0035 (t -2.7) | -0.04 | 2/10 |
| GS21 | `gxyz1rp` – gyzylf with riskless debt and a persistent y | 0.475 | 76% | 0.457 | +0.017 | 4 | 0.13 | 0.448 | 0.428 | 0.419 | 0.451 | 0.356 | -0.0035 (t -2.7) | -0.04 | 2/10 |
| GS21 | `gsbgj` – benchmark at $\gamma_x$ 0.1 | 0.070 | 96% | 0.070 | +0.000 | 0 | 0.00 | 0.061 | 0.029 | 0.030 | 0.061 | 0.054 | +0.0004 (t 0.9) | +0.00 | 6/10 |
| GS21 | `gxyz10g` – gxyz10 at $\gamma_x$ 0.1 | 0.770 | 25% | 0.707 | +0.051 | 7 | 0.27 | 0.703 | 0.722 | 0.647 | 0.724 | 0.195 | -0.0229 (t -6.5) | -0.18 | 0/10 |
| GS21 | `gxyz1g` – gyzylf at $\gamma_x$ 0.1 | 0.789 | 27% | 0.718 | +0.064 | 9 | 0.31 | 0.725 | 0.724 | 0.657 | 0.729 | 0.217 | -0.0051 (t -1.7) | -0.06 | 3/10 |
| GS21 | `gxyp10c` – the product with a leverage-level term, exposure 0.10 $\tilde{b}(\tilde{z}+2)$ | 0.774 | -86% | 0.763 | +0.006 | 1 | 0.09 | 0.745 | 0.727 | 0.720 | 0.767 | 0.674 | -0.0221 (t -4.1) | -0.17 | 0/10 |
| GS21 | `gxyt1` – three exposure types × three fixed leverage types (KP14's recipe) | 0.739 | 84% | 0.728 | +0.009 | 1 | 0.12 | 0.708 | 0.688 | 0.685 | 0.714 | 0.621 | -0.0060 (t -3.7) | -0.07 | 2/10 |
| GS21 | `gxyt1g` – the same at $\gamma_x$ 0.1 | 0.517 | 72% | 0.498 | +0.017 | 3 | 0.13 | 0.473 | 0.448 | 0.443 | 0.482 | 0.374 | -0.0089 (t -4.7) | -0.09 | 0/10 |
| BGN | `bry9t2` – two priced chains, nine types (independent ladders), total price 2.5 | 0.636 | 42% | 0.398 | +0.063 | 16 | 0.23 | 0.395 | 0.339 | 0.355 | 0.383 | 0.266 | +0.0124 (t 3.5) | +0.08 | 8/10 |
| BGN | `brych8` – brych0 with sign-switching types: six types, loadings ±0.01, ±0.035, ±0.07 | 0.510 | 36% | 0.349 | +0.069 | 20 | 0.23 | 0.360 | 0.274 | 0.277 | 0.334 | 0.183 | +0.0255 (t 4.9) | +0.12 | 9/10 |
| KP14 | `vyg25s` – vyg25 with sign-switching types: six types, loadings ±0.02, ±0.07, ±0.14 | 0.575 | 11% | 0.174 | +0.349 | 226 | 0.49 | 0.373 | 0.102 | 0.190 | 0.153 | 0.056 | +0.1810 (t 17.2) | +0.32 | 10/10 |
| KP14 | `vyg25u` – vyg25 without the level compensation: the type shows in value at $y = 0$ | 0.495 | 11% | 0.351 | +0.066 | 19 | 0.22 | 0.339 | 0.290 | 0.293 | 0.341 | 0.044 | -0.0018 (t -0.5) | -0.02 | 3/10 |
| GS21 | `gxy25s` – gyz25 with sign-switching types: six level-exposure types, ±0.015, ±0.05, ±0.1 | 0.619 | 52% | 0.554 | +0.061 | 11 | 0.27 | 0.549 | 0.541 | 0.506 | 0.560 | 0.318 | -0.0118 (t -4.4) | -0.10 | 2/10 |
| KP14 | `kpsgn` – the price of x switches sign with y: $\gamma_x(2\,\text{logistic}(-2y)-1)$ (conditioning room) | 0.143 | 42% | 0.075 | +0.050 | 90 | 0.10 | 0.060 | 0.018 | 0.045 | 0.064 | -0.002 | -0.0041 (t -0.9) | -0.01 | 1/3 |
| KP14 | `vyn6` – six priced states, 24 rank-6 types (vym3's construction at K = 6) | 0.421 | 15% | 0.300 | +0.056 | 19 | 0.19 | 0.287 | 0.260 | 0.275 | 0.286 | 0.059 | +0.0002 (t 0.1) | -0.01 | 4/10 |
| KP14 | `vym3s` – vyg25s's signed ladder over three priced states: 24 mean-zero types, mirrored alignments | 0.503 | 13% | 0.191 | +0.224 | 126 | 0.37 | 0.296 | 0.134 | 0.195 | 0.171 | 0.065 | +0.0998 (t 20.9) | +0.22 | 10/10 |
| KP14 | `vym6s` – vyg25s's signed ladder over six priced states: 24 mean-zero types, mirrored alignments, orthogonal off-$\gamma$ directions | 0.507 | 13% | 0.192 | +0.225 | 124 | 0.37 | 0.296 | 0.132 | 0.194 | 0.168 | 0.065 | +0.0987 (t 15.0) | +0.22 | 10/10 |
| GS21 | `gxy2s` – gxy25's signed ladder over two priced chains, cash-flow exposure only: twelve mean-zero types | 0.625 | 53% | 0.567 | +0.046 | 8 | 0.23 | 0.540 | 0.542 | 0.505 | 0.564 | 0.333 | -0.0246 (t -5.1) | -0.16 | 0/2 |
| GS21 | `gxyt1s` – gyytl with the exposure ladder mirrored: six signed exposure types × three fixed coupons, eighteen mean-zero types | 0.627 | 51% | 0.580 | +0.044 | 8 | 0.23 | 0.580 | 0.493 | 0.544 | 0.588 | 0.313 | -0.0074 (t -1.0) | -0.06 | 1/3 |
| BGN | `brych64` – brychs at bv_comp 0.4 | 0.612 | 29% | 0.519 | +0.059 | 12 | 0.25 | 0.497 | 0.468 | 0.465 | 0.497 | 0.177 | -0.0035 (t -0.3) | -0.01 | 2/5 |
| BGN | `brych66` – brychs at bv_comp 0.6 | 0.656 | 28% | 0.578 | +0.047 | 8 | 0.23 | 0.547 | 0.493 | 0.512 | 0.545 | 0.183 | +0.0013 (t 0.3) | +0.00 | 1/3 |
| BGN | `brych68` – brychs at bv_comp 0.8 | 0.684 | 28% | 0.590 | +0.044 | 7 | 0.23 | 0.577 | 0.501 | 0.468 | 0.568 | 0.189 | +0.0086 (t 1.0) | +0.06 | 2/3 |
| KP14 | `vyg25s6` – vyg25s at bv_comp 0.6 | 0.576 | 11% | 0.212 | +0.305 | 147 | 0.47 | 0.337 | 0.183 | 0.131 | 0.185 | 0.062 | +0.1424 (t 24.2) | +0.27 | 5/5 |
| KP14 | `vyg25s18` – vyg25s at bv_comp 1.8 | 0.574 | 11% | 0.209 | +0.312 | 167 | 0.47 | 0.387 | 0.106 | 0.172 | 0.193 | 0.062 | +0.1854 (t 8.7) | +0.33 | 5/5 |
| GS21 | `gxyib2` – gyyib at gs_bib −0.2 (the −1.0 and −0.5 solves diverged: the firm picks the coupon that makes investment near-free) | 0.327 | 96% | 0.326 | +0.001 | 0 | 0.03 | 0.319 | 0.298 | 0.293 | 0.322 | 0.314 | -0.0027 (t -11.0) | -0.04 | 0/10 |
| GS21 | `gxq36` – gsbase with the x loading hump-shaped in productivity: $1-0.35\,(z^2-1)/\sqrt{2}$ | 0.284 | 96% | 0.282 | +0.002 | 1 | 0.03 | 0.270 | 0.248 | 0.242 | 0.270 | 0.269 | +0.0002 (t 0.4) | +0.00 | 1/3 |
| GS21 | `gxq70` – gsbase with the x loading hump-shaped in productivity: $1-0.70\,(z^2-1)/\sqrt{2}$ | 0.278 | 95% | 0.269 | +0.008 | 3 | 0.07 | 0.261 | 0.211 | 0.201 | 0.261 | 0.259 | +0.0006 (t 1.8) | +0.01 | 2/3 |
| GS21 | `gxy25sq` – gyy25s at $\gamma_x$ 0.25, bv_comp 12.0: the level shifts kept, the y price cut | 0.307 | 87% | 0.305 | +0.002 | 1 | 0.03 | 0.290 | 0.262 | 0.239 | 0.294 | 0.263 | -0.0038 (t -8.2) | -0.05 | 0/3 |
| GS21 | `gxy25sh` – gyy25s at $\gamma_x$ 0.5, bv_comp 9.0: the level shifts kept, the y price cut | 0.351 | 71% | 0.341 | +0.009 | 3 | 0.08 | 0.328 | 0.309 | 0.286 | 0.336 | 0.246 | -0.0077 (t -2.5) | -0.07 | 0/3 |
| BGN | `brych618` – brychs at bv_comp 1.8 | 0.340 | 49% | 0.207 | +0.055 | 27 | 0.16 | 0.218 | 0.198 | 0.178 | 0.208 | 0.167 | +0.0064 (t 1.4) | +0.03 | 6/8 |
| BGN | `brych625` – brychs at bv_comp 2.5 | 0.449 | 36% | 0.250 | +0.163 | 67 | 0.33 | 0.341 | 0.286 | 0.223 | 0.246 | 0.159 | +0.0482 (t 3.7) | +0.15 | 9/10 |
| BGN | `brych630` – brychs at bv_comp 3.0 | 0.488 | 32% | 0.277 | +0.182 | 67 | 0.36 | 0.384 | 0.318 | 0.268 | 0.279 | 0.157 | +0.0626 (t 5.6) | +0.20 | 10/10 |
| BGN | `brych635` – brychs at bv_comp 3.5 | 0.508 | 31% | 0.294 | +0.189 | 67 | 0.38 | 0.403 | 0.315 | 0.309 | 0.303 | 0.156 | +0.0712 (t 5.8) | +0.22 | 10/10 |
| BGN | `brych640` – brychs at bv_comp 4.0 | 0.518 | 30% | 0.305 | +0.189 | 65 | 0.39 | 0.413 | 0.305 | 0.334 | 0.319 | 0.155 | +0.0590 (t 4.6) | +0.23 | 10/10 |
| BGN | `brych645` – brychs at bv_comp 4.5 | 0.525 | 29% | 0.281 | +0.224 | 81 | 0.42 | 0.422 | 0.316 | 0.345 | 0.315 | 0.154 | +0.0638 (t 2.5) | +0.21 | 4/4 |
| BGN | `brychm5` – brychs at bv_comp -0.5 | 0.367 | 35% | 0.254 | +0.080 | 35 | 0.21 | 0.229 | 0.202 | 0.207 | 0.228 | 0.127 | -0.0054 (t -0.9) | -0.01 | 6/10 |
| BGN | `bryvee` – V-shaped loading across a visible level ladder: level 0.6..1.4, $b = +0.07, +0.00, -0.07, +0.00, +0.07$ | 0.485 | 34% | 0.352 | +0.076 | 23 | 0.24 | 0.352 | 0.305 | 0.320 | 0.344 | 0.164 | +0.0071 (t 1.3) | +0.05 | 6/8 |
| GS21 | `gxyq110` – U-shaped exposure in the firm's own leverage: $+0.1\,(\tilde{b}^2-1)$, one type | 0.287 | -81% | 0.259 | +0.015 | 8 | 0.08 | 0.247 | 0.230 | 0.231 | 0.249 | 0.238 | -0.0021 (t -1.9) | -0.03 | 0/3 |
| GS21 | `gxyq120` – U-shaped exposure in the firm's own leverage: $+0.2\,(\tilde{b}^2-1)$, one type | 0.602 | -95% | 0.593 | +0.007 | 1 | 0.09 | 0.581 | 0.570 | 0.565 | 0.583 | 0.571 | -0.0020 (t -1.9) | -0.05 | 0/3 |
| GS21 | `gxyqh10` – hump-shaped exposure in the firm's own leverage: $-0.1\,(\tilde{b}^2-1)$, one type | 0.469 | -66% | 0.409 | +0.032 | 11 | 0.16 | 0.366 | 0.358 | 0.335 | 0.391 | 0.339 | -0.0259 (t -6.9) | -0.14 | 0/3 |
| KP14 | `vyg25sx2` – vyg25s with the loadings doubled: ±0.04, ±0.14, ±0.28 | 0.640 | 6% | 0.158 | +0.437 | 343 | 0.57 | 0.456 | 0.206 | 0.105 | 0.134 | 0.027 | +0.2452 (t 11.1) | +0.40 | 10/10 |
| KP14 | `vyg25sx3` – vyg25s with the loadings tripled: ±0.06, ±0.21, ±0.42 | 0.661 | 5% | 0.248 | +0.364 | 175 | 0.55 | 0.461 | 0.223 | 0.098 | 0.237 | 0.021 | +0.2177 (t 9.7) | +0.39 | 10/10 |
| GS21 | `gxyzz10` – U-shaped exposure in the firm's own productivity: $+0.1\,(z^2-1)$, one type | 0.621 | 43% | 0.356 | +0.224 | 60 | 0.45 | 0.498 | 0.455 | 0.312 | 0.489 | 0.252 | +0.0086 (t 0.7) | +0.02 | 3/5 |
| GS21 | `gxyzz20` – U-shaped exposure in the firm's own productivity: $+0.2\,(z^2-1)$, one type | 0.686 | 26% | 0.441 | +0.168 | 43 | 0.42 | 0.521 | 0.350 | 0.321 | 0.462 | 0.179 | +0.0589 (t 6.3) | +0.23 | 10/10 |
| GS21 | `gxyzzh10` – hump-shaped exposure in the firm's own productivity: $-0.1\,(z^2-1)$, one type | 0.639 | 25% | 0.364 | +0.250 | 70 | 0.49 | 0.480 | 0.325 | 0.255 | 0.400 | 0.152 | +0.0795 (t 4.6) | +0.25 | 10/10 |
| GS21 | `gxy25sc25` – gxy25s at bv_comp 2.5: strong level compensation | 0.712 | 33% | 0.542 | +0.168 | 31 | 0.46 | 0.635 | 0.585 | 0.630 | 0.655 | 0.231 | -0.0198 (t -2.8) | -0.14 | 1/6 |
| BGN | `bryhz2` – hump-shaped loading across a mild level ladder 0.85..1.15: $b = -0.14, +0.00, +0.14, +0.00, -0.14$ | 0.597 | 27% | 0.526 | +0.057 | 11 | 0.25 | 0.523 | 0.392 | 0.315 | 0.508 | 0.158 | +0.0150 (t 0.7) | +0.06 | 2/3 |
| BGN | `bryhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b = -0.21, +0.00, +0.21, +0.00, -0.21$ | 0.551 | 28% | 0.349 | +0.155 | 44 | 0.36 | 0.375 | 0.295 | 0.280 | 0.340 | 0.153 | +0.0345 (t 3.9) | +0.14 | 10/10 |
| BGN | `bryhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b = -0.28, +0.00, +0.28, +0.00, -0.28$ | 0.642 | 25% | 0.357 | +0.260 | 73 | 0.50 | 0.476 | 0.341 | 0.293 | 0.344 | 0.159 | +0.1186 (t 10.9) | +0.31 | 10/10 |
| KP14 | `vyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b = -0.21, +0.00, +0.21, +0.00, -0.21$ | 0.632 | 7% | 0.207 | +0.377 | 197 | 0.54 | 0.431 | 0.217 | 0.136 | 0.178 | 0.031 | +0.2123 (t 12.5) | +0.37 | 10/10 |
| KP14 | `vyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b = -0.28, +0.00, +0.28, +0.00, -0.28$ | 0.659 | 5% | 0.298 | +0.318 | 123 | 0.53 | 0.480 | 0.309 | 0.199 | 0.278 | 0.022 | +0.1664 (t 10.6) | +0.36 | 10/10 |
| GS21 | `gryhx3` – hump-shaped loading across a mild level ladder 0.8..1.2 (gs_level): $b = -0.21, +0.00, +0.21, +0.00, -0.21$ | 0.823 | 26% | 0.706 | +0.112 | 17 | 0.41 | 0.766 | 0.737 | 0.725 | 0.764 | 0.210 | +0.0023 (t 0.9) | +0.04 | 7/10 |
| BGN | `bryq2` – two chains, the second loading quadratic in the first: $b_2 = -14\,(b_1^2 - b_1^2)$, $b_1 = \pm 0.01, \pm 0.035, \pm 0.07$ | 0.460 | 37% | 0.303 | +0.063 | 21 | 0.20 | 0.307 | 0.240 | 0.260 | 0.285 | 0.172 | +0.0223 (t 18.1) | +0.11 | 3/3 |
| BGN | `bryq2x` – two chains, the second loading quadratic in the first: $b_2 = -7\,(b_1^2 - b_1^2)$, $b_1 = \pm 0.02, \pm 0.07, \pm 0.14$ | 0.523 | 30% | 0.375 | +0.070 | 19 | 0.24 | 0.359 | 0.321 | 0.308 | 0.366 | 0.158 | -0.0075 (t -1.2) | -0.05 | 1/3 |
| BGN | `bryq2k` – two chains, the second loading quadratic in the first: $b_2 = -28\,(b_1^2 - b_1^2)$, $b_1 = \pm 0.01, \pm 0.035, \pm 0.07$ | 0.470 | 34% | 0.315 | +0.071 | 22 | 0.22 | 0.304 | 0.265 | 0.261 | 0.302 | 0.159 | +0.0026 (t 0.9) | +0.01 | 6/10 |
| KP14 | `vyg25su` – vyg25s without level compensation (bv_comp 0): ±0.02, ±0.07, ±0.14 | 0.575 | 11% | 0.290 | +0.230 | 84 | 0.43 | 0.395 | 0.270 | 0.066 | 0.274 | 0.056 | +0.1162 (t 11.1) | +0.28 | 10/10 |
| KP14 | `vyg25sux2` – vyg25sx2 without level compensation (bv_comp 0): ±0.04, ±0.14, ±0.28 | 0.640 | 6% | 0.185 | +0.411 | 500 | 0.55 | 0.482 | 0.284 | 0.088 | 0.229 | 0.027 | +0.1914 (t 6.5) | +0.37 | 10/10 |
| KP14 | `vyg25x2` – vyg25 (positive ladder) with the loadings x2: 0.04, 0.14, 0.28 | 0.468 | 7% | 0.335 | +0.074 | 22 | 0.23 | – | – | – | – | – | – | – | – |
| KP14 | `vyg25x3` – vyg25 (positive ladder) with the loadings x3: 0.06, 0.21, 0.42 | 0.475 | 6% | 0.358 | +0.063 | 17 | 0.22 | – | – | – | – | – | – | – | – |
| KP14 | `vyg35x2` – vyg25x2 (positive ladder x2) with the $y$ price raised to 3.5 | 0.531 | 5% | 0.373 | +0.079 | 21 | 0.26 | – | – | – | – | – | – | – | – |
| KP14 | `vyg25x2c` – vyg25x2 (positive ladder x2) with the level compensation strengthened to 2.5 | 0.489 | 7% | 0.359 | +0.074 | 20 | 0.24 | – | – | – | – | – | – | – | – |
| KP14 | `vygo` – the $y$ exposure on the profitability deviation, $b_Y\,(e-1)$: three types 0.1 / 0.25 / 0.45 | 0.477 | 19% | 0.267 | +0.038 | 14 | 0.15 | 0.250 | 0.228 | 0.254 | 0.251 | 0.088 | -0.0048 (t -1.9) | -0.05 | 0/3 |
| KP14 | `vygxl` – vygo with a visible level ladder, type-level 0.6 / 1.0 / 1.6 | 0.491 | 17% | 0.269 | +0.046 | 17 | 0.16 | 0.266 | 0.254 | 0.236 | 0.256 | 0.077 | +0.0049 (t 1.1) | +0.03 | 2/3 |
| KP14 | `vygesm` – exposure on the profitability deviation, mirrored types ±0.05 / 0.1 / 0.15 with a level ladder 0.5..2.0: a product of two mean-zero factors | 0.369 | 20% | 0.232 | +0.022 | 9 | 0.10 | 0.209 | 0.203 | 0.189 | 0.217 | 0.074 | -0.0083 (t -1.4) | -0.04 | 1/3 |
| KP14 | `vygelb` – bounded exposure (cap 2) on the profitability deviation, mirrored types ±0.1 / 0.25 / 0.45 with a level ladder 0.5..2.0 | 0.482 | 16% | 0.228 | +0.058 | 26 | 0.17 | 0.219 | 0.199 | 0.203 | 0.215 | 0.074 | +0.0024 (t 1.4) | +0.02 | 7/10 |
| KP14 | `vygelb7` – bounded exposure (cap 2), mirrored types ±0.2 / 0.45 / 0.7 with a level ladder 0.5..2.0 | 0.535 | 14% | 0.231 | +0.085 | 37 | 0.22 | 0.225 | 0.202 | 0.208 | 0.218 | 0.073 | +0.0049 (t 2.1) | +0.03 | 8/10 |
| KP14 | `vygelb7g` – bounded exposure (cap 2), mirrored types ±0.2 / 0.45 / 0.7, level ladder 0.5..2.0, $\gamma_x$ 4.0 | 0.677 | 11% | 0.235 | +0.120 | 51 | 0.27 | 0.238 | 0.208 | 0.208 | 0.222 | 0.073 | +0.0135 (t 4.2) | +0.07 | 10/10 |
| KP14 | `vygelbw` – bounded exposure (cap 2), mirrored types ±0.3 / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_x$ 4.0 | 0.710 | 11% | 0.243 | +0.138 | 56 | 0.29 | 0.254 | 0.216 | 0.216 | 0.228 | 0.073 | +0.0215 (t 4.3) | +0.10 | 10/10 |
| KP14 | `vygelbx` – bounded exposure (cap 2), mirrored types ±0.5 / 1.2 / 2.0, level ladder 0.5..2.0, $\gamma_x$ 4.0 | 0.727 | 11% | 0.254 | +0.150 | 58 | 0.31 | 0.273 | 0.227 | 0.216 | 0.241 | 0.073 | +0.0297 (t 4.2) | +0.12 | 10/10 |
| KP14 | `vygelbc` – bounded exposure (cap 2.5), mirrored types ±0.3 / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_x$ 4.0 | 0.752 | 9% | 0.244 | +0.170 | 69 | 0.33 | 0.272 | 0.220 | 0.219 | 0.231 | 0.068 | +0.0369 (t 6.6) | +0.14 | 10/10 |
| KP14 | `vygelbc6` – bounded exposure (cap 2.5), mirrored types ±0.3 / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_x$ 6.0 | 0.892 | 8% | 0.249 | +0.201 | 80 | 0.37 | 0.290 | 0.230 | 0.222 | 0.237 | 0.069 | +0.0487 (t 5.8) | +0.16 | 10/10 |
| KP14 | `vygelbcs` – bounded exposure (cap 2.5), mirrored types ±0.3 / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_x$ 4.0, $\sigma$, 0.3 | 0.858 | 7% | 0.297 | +0.256 | 90 | 0.46 | 0.388 | 0.251 | 0.288 | 0.282 | 0.052 | +0.0913 (t 12.3) | +0.25 | 10/10 |
| GS21 | `gryls115`[†] – industry sign × leverage: two industries (gs_level 0.8 / 1.2) × fixed coupons 0.15 / 0.40 / 0.70, $b$ = sign × 0.15 × coupon/0.4 | 1.190 | 66% | 1.183 | +0.006 | 1 | 0.12 | – | – | – | – | – | – | – | – |
| GS21 | `gryzz2` – type sign × productivity $\tilde{z}$: mirrored types ±0.15 / 0.3 / 0.5, gs_level ladder 0.5..2.0, no compensation | 0.847 | 17% | 0.603 | +0.208 | 37 | 0.54 | 0.678 | 0.536 | 0.488 | 0.619 | 0.151 | +0.0590 (t 11.5) | +0.27 | 10/10 |
| GS21 | `gryzz1` – type sign × productivity $\tilde{z}$: mirrored types ±0.1 / 0.2 / 0.35, gs_level ladder 0.5..2.0, no compensation | 0.844 | 25% | 0.530 | +0.278 | 60 | 0.60 | 0.658 | 0.560 | 0.431 | 0.605 | 0.210 | +0.0533 (t 6.1) | +0.24 | 10/10 |
| BGN | `brychsx2` – brychs with the loadings x 2: ±0.02, ±0.07, ±0.14 | 0.496 | 32% | 0.344 | +0.089 | 26 | 0.26 | 0.347 | 0.308 | 0.279 | 0.333 | 0.161 | +0.0072 (t 1.1) | +0.04 | 7/10 |
| BGN | `brychsx3` – brychs with the loadings x 3: ±0.03, ±0.105, ±0.21 | 0.533 | 31% | 0.384 | +0.081 | 21 | 0.26 | 0.377 | 0.353 | 0.321 | 0.370 | 0.164 | +0.0024 (t 0.5) | +0.01 | 6/10 |
| GS21 | `gryzz5x2` – gryz25s with the loadings x 2: ±0.03, ±0.1, ±0.2 | 0.817 | 41% | 0.721 | +0.093 | 13 | 0.38 | 0.814 | 0.777 | 0.796 | 0.814 | 0.344 | +0.0005 (t 0.1) | +0.01 | 2/3 |
| KP14 | `kpgsm` – $\gamma_x \times$ logistic($y$), 1 to 3 | 0.390 | 65% | 0.373 | +0.005 | 1 | 0.06 | 0.358 | 0.348 | 0.340 | 0.363 | 0.254 | -0.0053 (t -4.0) | -0.05 | 1/10 |
| KP14 | `vyx` – three types on y, price 1.8 | 0.421 | 14% | 0.321 | +0.055 | 17 | 0.19 | 0.313 | 0.284 | 0.302 | 0.310 | 0.054 | +0.0009 (t 0.3) | +0.02 | 7/10 |
| KP14 | `vyj25` – the same with a priced crash in y | 0.554 | 10% | 0.323 | +0.104 | 33 | 0.28 | 0.315 | 0.270 | 0.292 | 0.307 | 0.045 | +0.0080 (t 2.2) | +0.06 | 8/10 |
| KP14 | `vym2t3` – two priced states, three loading vectors | 0.475 | 12% | 0.326 | +0.090 | 28 | 0.26 | 0.317 | 0.280 | 0.301 | 0.308 | 0.049 | +0.0075 (t 2.1) | +0.05 | 8/10 |
| KP14 | `vy9t2f` – two priced states, nine types, each state priced 2.5 | 0.582 | 6% | 0.415 | +0.077 | 19 | 0.26 | 0.416 | 0.314 | 0.339 | 0.401 | 0.018 | +0.0149 (t 4.2) | +0.10 | 9/10 |
| KP14 | `vym3t3` – three priced states, three types | 0.443 | 14% | 0.285 | +0.087 | 30 | 0.24 | 0.279 | 0.246 | 0.266 | 0.273 | 0.061 | +0.0036 (t 0.8) | +0.02 | 6/10 |
| KP14 | `vym3` – three priced states, twenty rank-3 types | 0.418 | 15% | 0.298 | +0.056 | 19 | 0.19 | 0.286 | 0.259 | 0.278 | 0.287 | 0.059 | -0.0032 (t -1.7) | -0.02 | 5/10 |
| RF | `kfh_tanh_1` – K=1, bounded (tanh) loadings on observed characteristics | 0.592 | 8% | 0.592 | +0.001 | 0 | 0.03 | 0.579 | 0.578 | 0.570 | 0.588 | 0.041 | -0.0105 (t -1.9) | -0.10 | 0/5 |
| RF | `kfh_tanh_4` – K=4 | 0.584 | 9% | 0.574 | +0.007 | 1 | 0.09 | 0.541 | 0.564 | 0.544 | 0.560 | 0.041 | -0.0232 (t -2.9) | -0.16 | 0/5 |
| RF | `kfh_tanh_16` – K=16 | 0.469 | 11% | 0.378 | +0.044 | 12 | 0.19 | 0.346 | 0.357 | 0.345 | 0.359 | 0.041 | -0.0152 (t -1.4) | -0.10 | 1/5 |
| RF | `kfh_sq_1` – K=1, pure curvature (squares) of observed characteristics | 0.600 | 8% | 0.080 | +0.519 | 647 | 0.59 | 0.582 | 0.032 | 0.046 | 0.059 | 0.041 | +0.5161 (t 57.3) | +0.58 | 5/5 |
| RF | `kfh_sq_4` – K=4 | 0.601 | 8% | 0.097 | +0.501 | 515 | 0.59 | 0.573 | 0.042 | 0.458 | 0.074 | 0.041 | +0.1155 (t 22.6) | +0.35 | 5/5 |
| RF | `kfh_sq_16` – K=16 | 0.601 | 8% | 0.076 | +0.523 | 687 | 0.59 | 0.571 | 0.036 | 0.429 | 0.052 | 0.041 | +0.1429 (t 14.3) | +0.38 | 5/5 |
| RF | `kfo_sq_1` – K=1, squares, directions orthog

- `g0235df`: the same $g$ with switch probabilities 1/6 and 1/12: stress two thirds of months, mean spell 12 months.
- `bvy25`: `bgnbase` plus $y$; type $f$'s project cash flows are $\theta_f C_j e^{b_f y_t}$, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated. Exposure is $b_f$ times the firm's mix of assets in place and growth option.
- `bvych41`: `bvych0` with $y$ on a 41-node chain instead of 21.
- `bvy9t2`: `bvych0` with a second independent 21-node chain and nine types, a pair $(f_1, f_2)$ drawn independently from $\{0.01, 0.035, 0.07\}$ for each chain, cash flows $C_j\, e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each: the nested nine-type design of `vy9t2` in BGN.
- `bvyc0`: `bvy25` with $\theta_f = 1$, so the type is visible in value.
- `bvyj25`: `bvych0` (group 3b) plus a crash: each month, with probability 1% whatever $y_t$, $y_{t+1}$ goes to the lowest node; otherwise the Tauchen transition. The kernel weights the crash transition by $e^2$, so its risk-neutral probability is about 7.4% a month before renormalization.
- `bvysg`: `bgnbase` plus $y$; type $f$'s project cash flows are $C_j e^{b_f y_t}$, $b_f \in \{-0.07, 0, +0.07\}$, not level-compensated.
- `bvychs`: `bvych0` with the ladder mirrored: six types of equal share, $b_f \in \{-0.07, -0.035, -0.01, 0.01, 0.035, 0.07\}$, level-compensated (bv_comp 1.2, $\theta_f$ 1 to 1.60 with $b = -0.07$ the reference), 21-node chain, price 2.5.
- `bvychs18`, `bvychs25`, `bvychs30`, `bvychs35`, `bvychsn5`: `bvychs` at bv_comp 1.8 / 2.5 / 3.0 / 3.5 / −0.5. Ten seeds each (bvychs18 eight). bgn_vy's SDF step had its K4 pair sum rewritten as a blocked sum (commit 78961c4) after these ran out of memory at 36 GB; the arithmetic is unchanged to $3 \times 10^{-15}$.
- `bvyvee`: five types of equal share on a visible level ladder, $\theta_f$ = 0.6 / 0.8 / 1.0 / 1.2 / 1.4 multiplying the cash-flow level, exposure V-shaped across it, $b_f$ = +0.07, 0, −0.07, 0, +0.07; bv_comp 1.0, price 2.5. Eight seeds; two stopped. `bvyhump`, its mirror, was solved and not simulated.
- `bvychsx2`, `bvychsx3`: `bvychs` with the loadings ×2 (±0.02, ±0.07, ±0.14) and ×3 (±0.03, ±0.105, ±0.21). Ten seeds each.
- `bvychs40`, `bvychs45`: `bvychs` at bv_comp 4.0 / 4.5. Ten seeds; four seeds, stopped.
- `bvyhx2`, `bvyhx3`, `bvyhx4`: five types on a mild visible level ladder ($\theta_f$ 0.85–1.15 / 0.80–1.20 / 0.80–1.20) with the loading hump-shaped across it, $b_f = -c, 0, +c, 0, -c$, $c$ = 0.14 / 0.21 / 0.28; bv_comp 1, price 2.5. Three seeds (stopped, room 9–13%); ten; ten.
- `bvyu4`, `bvyu5`, `bvyu6`, `bvyu5r`, `bvyu6r`: the hump ladder's chassis with a U-shaped loading, $+c$ at the two ends, $c$ = 0.28 / 0.35 / 0.42; `bvyu5` failed the gate (the small exposed type is the compensation's reference, $J^* = 0$), `bvyu6` was not built, the `r` versions list the unexposed level-1.0 type first. Ten seeds each.
- `bvyq2`, `bvyq2x`, `bvyq2k`: `bvychs`'s six types with a second priced chain (both at $2.5/\sqrt{2}$) whose loading is quadratic in the first, $b_2 = \kappa\,(b_1^2 - \overline{b_1^2})$: $\kappa = -14$ on ±0.01/0.035/0.07; −7 on ±0.02/0.07/0.14; −28 on ±0.01/0.035/0.07. Three seeds each (stopped); ten.

**GS21**

- `g28`: `gsbase` with $\gamma_x \to \gamma(x_t) = \text{clip}\big(0.5 - 0.28\,x_t/\text{sd}(x),\ 0.05,\ 1.0\big)$, countercyclical.
- `bx7`: `gsbase` with $\gamma_x \to g_s\gamma_x$, $g \in \{0.6, 3.0\}$ on a two-state Markov regime, and five types with output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$.
- `gx7`: `gsbase` with five types, output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$, and $\gamma(x_t) = \text{clip}\big(0.5 - 0.4\,x_t/\text{sd}(x),\ 0.05,\ 1.5\big)$. A firm's exposure to the one priced shock is its type, which size and profitability reveal.
- `gvy1`: `gsbase` plus $y$ (9-node chain over $\pm 2.5$ sd), one type; output $e^{x_t + z_{it} + 0.05\,y_t} k_{it}$, a level exposure.
- `gvy1j`: `gvy1` with $\gamma_v = 0$ and the crash, on a chain over $\pm 3.5$ sd.
- `gvy25`: `gsbase` plus $y$ (9-node chain); output $e^{x_t + z_{it} + b_f y_t + a_f} k_{it}$, $b_f \in \{0.015, 0.05, 0.10\}$, with $a_f$ the level compensation.
- `gvy25s`: `gvy25` with the ladder mirrored: six types of equal share, $b_f \in \{-0.10, -0.05, -0.015, 0.015, 0.05, 0.10\}$, level-compensated (bv_comp 1.2) at price 2.5.
- `gvys25`: `gsbase` plus $y$, entering through an operating cost: $0.2\,e^{c_f y_t}$ per unit capital is subtracted from output, $c_f \in \{+0.4, 0, -0.4\}$; output itself does not load on $y$.
- `gvyp05, gvyp075, gvyp15, gvyp20`: `gvyp10` with the coefficient 0.10 replaced by 0.05, 0.075, 0.15 and 0.20.
- `gvyp10x`: `gvyp10` with output $e^{(1+(\tilde{z}_{it}^2-1)/\sqrt{2})\,x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$: the loading on $x$ is also nonlinear in productivity.
- `gvyp10j`: `gvyp10` (group 3b) plus the same priced crash on its 9-node chain.
- `gvypk2`: `gvyp10` with a second independent chain (5 nodes each, 25 joint); two types of equal share with output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$ and $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t^{(2)}} k_{it}$.
- `gvyzl`: `gvyz10` with three leverage types: the coupon the firm refinances to is fixed at 0.158 / 0.421 / 0.684 per unit capital, equal shares drawn independently of productivity; refinancing opportunities (1% a month) and dilution at investment as in GS21, so the realized coupon drifts below the type's.
- `gvyzlf`: `gvyzl` with the capital structure fixed for life: no refinancing, new debt issued pari passu at investment (debt-to-equity 0.25 / 1.0 / 3.25).
- `gvyzlw`: `gvyzlf` with coupons 0.05 / 0.45 / 0.85; the high type is distressed (premium 16% a year, default rate 1.4%).
- `gvyzl5`: `gvyzlf` with five types, coupons 0.10 to 0.70.
- `gvyzlb`: `gvyzlf` with the exposure doubled to $0.20\,\tilde{z}\,y$.
- `gvyzlr`: `gvyzlf` with riskless debt: the debt claim is worth its riskless coupon value in every state (creditors made whole on default).
- `gvyzlp`: `gvyzlf` with $\kappa_y = 0.10$ a year instead of 0.35.
- `gvyzlrp`: both.

- `gsbg1`: the GS21 benchmark with $\gamma_x = 0.1$ instead of 0.5 (`gs_vy` at one $y$ node and no exposure, which is `gsbase`'s solver).
- `gvyz10g`: `gvyz10` at $\gamma_x = 0.1$.
- `gvyzlg`: `gvyzlf` at $\gamma_x = 0.1$.
- `gvyp10c`: `gvyp10` with a leverage-level term, output $e^{x+z+0.10\,\tilde{b}(\tilde{z}+2)\,y}$: a levered firm's exposure keeps leverage's sign for every firm with $\tilde{z} > -2$. The mean firm ($\tilde{b} < 0$) is a hedge against $y$, so the market's premium is negative.
- `gvytl`: KP14's 3b recipe in GS21: three latent exposure types, output $e^{x+z+b_f y+a_f}$ with $b_f \in \{0.015, 0.05, 0.10\}$ and $a_f$ the level compensation, times three fixed leverage types (coupons 0.158 / 0.421 / 0.684, no refinancing), nine types of equal share.
- `gvytlg`: `gvytl` at $\gamma_x = 0.1$ (benchmark `gsbg1`).
- `gxq35`, `gxq70`: `gsbase` (no $y$) with the productivity loading $1 + c\,(\tilde{z}^2 - 1)/\sqrt{2}$, $c = -0.35$ / $-0.70$. Ten oracle seeds, three estimator seeds; stopped.
- `gvy25sq`, `gvy25sh`: `gvy25s` at $\gamma_v$ 0.25 / 0.5 with bv_comp 12 / 9 (the types' level shifts kept, the $y$ price cut). Ten oracle seeds, three estimator seeds; stopped.
- `gvyql10`, `gvyql20`, `gvyqh10`: one type on `gvy25s`'s chassis with the $y$ exposure $c\,(\tilde{b}^2 - 1)$, $\tilde{b}$ the standardized coupon, $c = +0.1$ / $+0.2$ / $-0.1$. Three seeds each; stopped.
- `gvy25sx2`: `gvy25s` with the loadings $\times 2$ ($\pm 0.03$, $\pm 0.10$, $\pm 0.20$). Five oracle seeds, three estimator seeds; stopped. `gvy25sx3`: the $b_f = +0.045$ type's solve oscillated to the sweep cap; not run.
- `gvyzz10`, `gvyzz20`, `gvyzzh10`: one type on `gvy25s`'s chassis with the $y$ exposure $c\,(\tilde{z}^2 - 1)$, $\tilde{z}$ the standardized idiosyncratic productivity, $c = +0.1$ / $+0.2$ / $-0.1$ (`gs_bzz`, new). Five (stopped), ten and ten seeds.
- `gvyhx3`, `gvyhx4`, `gvyu4`: the size ladder in GS21 through the new gs_level (a per-type production-level multiplier; 1.0 leaves every earlier economy unchanged): five types at 0.8–1.2, bv_comp 1.0, NY 9, y_mult 2.5, price 2.5, one solve per type; the hump $\pm 0.21$ / $\pm 0.28$ and the U $+0.28$. gvyhx3 ten seeds (no gap), gvyhx4 solved and not pursued, gvyu4 stopped at the oracle.
- `gvylu30`, `gvylu30c`: the exposure U-shaped in leverage on `gvyzlf`'s fixed-coupon chassis (coupons 0.15 / 0.40 / 0.70 for life), production exposure gs_bv (0.3, 0, 0.3), without and with the $y = 0$ value compensation; both stopped at the oracle.
- `gvysz1`, `gvysz2`: the industry sign times the firm's standardized productivity (gs_bzl 0, gs_bzi 1), six mirrored types $\pm 0.1/0.2/0.35$ and $\pm 0.15/0.3/0.5$ on the gs_level ladder 0.5–2.0, bv_comp 0, price 2.5, six solves each; ten seeds each.
- `gvysl15`, `gvysl20`: the industry sign times financial leverage, two industries (gs_level 0.8 / 1.2) times three fixed coupons, exposure sign $\times c \times$ coupon/0.40 at $c = 0.15$ / 0.20, bv_comp 1.0; gvysl15 stopped at the oracle, gvysl20 solved and not run.
- `gvy25sc25`: `gvy25s` at bv_comp 2.5. Six seeds; stopped. `gvy25sc32`: one type's solve cycle-capped; not run.

**KP14**

- `kpgam`: `kpbase` with $\gamma_x \to \gamma_x\, g(y_t)$, $g(y) = 1 + 2/(1 + e^{2y})$, rising from 1 when $y$ is high to 3 when it is low; $y$ is unpriced ($\gamma_v = 0$) and no firm is exposed to it.

- `vyx`: `kpbase` plus $y$ with $\gamma_v = 1.8$; type $f$'s project cash flows are $\theta_f e^{b_f y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated.

- `vyj25`: `vyg25` (group 3b) plus a Poisson jump of $y$ to $-y_{\max} = -3.5$ at intensity 0.12 a year (1% a month) under $P$ and $0.12\, e^2 = 0.89$ a year under $Q$.

- `vym3t3`: `vyg25` with $y \in \mathbb{R}^3$, three independent OU states; type $f$'s cash flows load $e^{b_f' y_t}$ with three vectors $b_f \in \mathbb{R}^3$ spanning the three states.

- `vym2t3`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; three types with loading vectors $(0.02, 0)$, $(0, 0.07)$, $(0.099, 0.099)$, price $2.5/\sqrt{2}$ on each: three sorts in three directions, which prices types 1 and 2 at an effective 1.77 instead of 2.5. The group-4 cell until October 21, 2026.

- `vy9t2f`: `vy9t2` with each state priced at 2.5 instead of $2.5/\sqrt{2}$; nests `vyg25` exactly.

- `vyg25s`: `vyg25` with the ladder mirrored: six types of equal share, $b_f \in \{-0.14, -0.07, -0.02, 0.02, 0.07, 0.14\}$, level-compensated as vyg25 (bv_comp 1.2, $\theta_f$ 1 to 3.36 with $b = -0.14$ the reference), price 2.5.

- `vyg25u`: `vyg25` with bv_comp 0: no level compensation, so the type shows in value at $y = 0$. The returns are vyg25's exactly ($\mathrm{SR}_{\max}$ equal to every digit); only the characteristics change.

- `vym3`: the same three states with twenty types, $b_f \in \mathbb{R}^3$ of rank 3.

What they add to the main results:

- **The chain's resolution is not what separates `bvych0` from `bvy25`**: on 41 nodes the gap is +0.009 against +0.008 on 21, and the difference from the Gaussian economy's +0.015 (sd 0.016) is within noise.

- **A level exposure of GS21 output to $y$** (`gvy1`, `gvy1j`) has no room at all, as every other level wiring in GS21; the productivity exposure (`gvyz10`) has 5% and the product with the coupon 43%.

- **The crash on an interaction economy** leaves room and gap at the control's in BGN (+0.007 against +0.008) and GS21 (+0.030 against +0.020, within noise) and halves KP14's (+0.008 against +0.014): the crash months are a state the ranks-times-state linear benchmark reads as well as DKKM does.

- **The GS21 product ladder**: room is hump-shaped in the product's size (45 / 111 / 47 / 22 / 6% of the linear ceiling at 0.05 / 0.075 / 0.10 / 0.15 / 0.20) while the gap is +0.019 / +0.077 / +0.020 / +0.078 / +0.022; above 0.10 the market becomes a hedge. A second nonlinear loading on $x$ (`gvyp10x`) gives +0.049: a second interaction factor adds. `bvy25`, the Gaussian-$y$ version of `bvych0`, has twice its gap at the same room, and the visibility ladder (`bvy25`, `bvyc0`, and `bvyc10` at `bv_comp` 1.0: +0.015 / +0.010 / +0.010) is flat within noise.

- **Two apparent failures of the mechanism are estimation cost, not structure.** `vyx` is `vyg25` at a lower price of $y$: its room (0.054) is about what DKKM gives up to estimation relative to the linear model (0.053), so the gap is zero, and the higher price lifts the room more than the cost. `bvysg` has twice `bvy25`'s room but DKKM's estimation loss is two and a half times larger (0.16 against 0.06): with signed exposures the nonlinear part of the tangency portfolio is a

direct bet on the persistent state $y$, whose sample mean a 360-month window barely pins down. `bvychs` (appendix), the signed ladder with six types and the level compensation, has a positive gap of +0.026 at room 20%, so the loss in `bvysg` came with the visible type, as in `vyg25u`. What matters is room net of DKKM's estimation cost. Four more GS21 wirings (investment cost, floating coupon, a larger option, state-dependent recovery) were stopped at the oracle stage with room below +0.01, and GS21's default channel is dormant at every calibration tried.

- **Knobs that are not mechanisms**: the speed of the priced state ($\kappa$ 0.15 / 0.35 / 0.70 in KP14: +0.007 / +0.014 / +0.009, a mild hump within noise), the visibility of the type, the number of types (`vym3` against `vym3t3`), the sample size (`vym3` at $N = 1000$: room up, gap flat), and the ridge grid (DKKM +0.0001). The price of the interacted component is the one scaling lever (`vyx` to `vyg25`).

- `vyg25sx2`, `vyg25sx3`: `vyg25s` with the loadings ×2 (±0.04, ±0.14, ±0.28; $\theta_f$ 1 to 11.2) and ×3 (±0.06, ±0.21, ±0.42; $\theta_f$ 1 to 37). Ten seeds each.

- `vyg25su`, `vyg25sux2`: `vyg25s` and `vyg25sx2` at bv_comp 0 (the parents' returns; the type revealed by value). Ten seeds each.

- `vyge`, `vygel`, `vygels`, `vygesm`: the $y$ exposure on the profitability deviation (`bv_on_eps`, cash flow $xK^{\alpha}\theta u[1+(\varepsilon-1)e^{b_f y}]$): three types 0.1 / 0.25 / 0.45 with $\theta_f = 1$; the same with `type_level` 0.6 / 1.0 / 1.6; six mirrored types ±0.1/0.25/0.45 on a level ladder 0.5–2.0 (solved, not run: negative unit values for $b_f \leq -0.25$); ±0.05/0.1/0.15 on the same ladder. Three seeds each; stopped at the oracle.

- `vygelb`, `vygelb7`, `vygelb7g`, `vygelbw`, `vygelbx`, `vygelbc`, `vygelbc6`, `vygelbcs`: the bounded exposure on the profitability deviation (`bv_eps_cap` $c$: $h_b(y) = c/(1+(c-1)e^{-cby/(c-1)})$ in place of $e^{b_f y}$), six mirrored types on the level ladder 0.5–2.0, bv_comp 1.2: ±0.1/0.25/0.45 (cap 2, $\gamma_v$ 2.5); ±0.2/0.45/0.7 ($\gamma_v$ 2.5 and 4.0); ±0.3/0.7/1.2 ($\gamma_v$ 4.0; cap 2 and 2.5); ±0.5/1.2/2.0 ($\gamma_v$ 4.0); cap 2.5 at $\gamma_v$ 6.0 or $\sigma_\varepsilon$ 0.3. Ten seeds each.

- `vyhx3`, `vyhx4`, `vyu4`, `vyu4r`, `vyu5r`: BGN's size ladder in KP14 through type_level: five types at 0.8–1.2, bv_comp 1.0, price 2.5; the hump ±0.21 / ±0.28 and the U +0.28 (vyu4r / vyu5r with the unexposed level-1.0 type first, +0.28 / +0.35). Ten seeds each.

- `vyg25x2`, `vyg25x3`, `vyg35x2`, `vyg25x2c`: `vyg25` with the loadings ×2 / ×3, and ×2 at $\gamma_v$ 3.5 / bv_comp 2.5. Three to six oracle seeds; stopped.

### Two priced states

A second independent priced state, with the total price held at the one-state value ($2.5/\sqrt{2}$ on each). In BGN and GS21 three types span the two states with loading vectors in the pattern $(1,0)$, $(0,1)$, $(1,1)/\sqrt{2}$, at the 3b exposure sizes; in GS21 each type carries the 0.10 product exposure to its state. In KP14 the cell is the nested extension of 3b: every firm draws a type for each state independently from the 3b ladder $(0.02, 0.07, 0.14)$, nine types in all, so exposure to each state is its own type times leverage and zeroing the second ladder recovers `vyg25`. The three-direction KP14 version (`vym2t3`), the nested version with each state priced at the full 2.5 (`vy9t2f`) and the nested BGN version (`bvy9t2`) are in the appendix.

**Verdict.** More factors are a lever only where the second state brings a second interaction. In BGN the second chain raises room by a quarter and doubles the gap (+0.015, ten of ten seeds). In GS21 it raises the product economy's gap from +0.020 to +0.056 (nineteen of twenty seeds), as the two-type

Table 10: Group 4. Larger factor models: two priced states. Two independent priced states at total price 2.5 ($2.5/\sqrt{2}$ each). BGN and GS21: three types with loading vectors (1,0), (0,1), $(1,1)/\sqrt{2}$. KP14: nine types, an independent draw from the 3b ladder for each state (the three-direction version and the full-price version are in the appendix).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvyk2` – two priced chains, three loading vectors | 0.639 | 36% | 0.393 | +0.071 | 18 | 0.25 | 0.394 | 0.332 | 0.353 | 0.379 | 0.228 | +0.0152 (t 4.3) | +0.10 | 10/10 |
| GS21 | `gvypk3` – two priced chains, three product types | 0.612 | 8% | 0.349 | +0.226 | 65 | 0.45 | 0.452 | 0.242 | 0.300 | 0.398 | 0.035 | +0.0515 (t 3.4) | +0.18 | 9/10 |
| KP14 | `vy9t2` – two priced states, nine types (independent ladders), total price 2.5 | 0.506 | 9% | 0.387 | +0.056 | 15 | 0.21 | 0.378 | 0.326 | 0.337 | 0.373 | 0.031 | +0.0048 (t 2.3) | +0.04 | 7/10 |

version did (appendix). In KP14 it does not help however it is wired: the nested nine-type cell lowers room from 24% to 15% of the linear ceiling and the gap from +0.015 to +0.005; the three-direction version (`vym2t3`) keeps the room (28%) but halves the gap (+0.008); pricing each state at the full 2.5 (`vy9t2f`) only returns the gap to 3b's (+0.015, nine of ten seeds) at lower room (19%). The reason is in the ceilings: with an independent second type a firm's total exposure is the sum of two ladder draws, six evenly spread values that size and momentum proxy almost linearly, so the linear ceiling rises more than the nonlinear one (0.346 to 0.387). Both states act through the same channel, the growth-option elasticity, so the second state adds premium and a factor but not a second interaction. BGN's nine-type version (`bvy9t2`, +0.012, room 16%) sits between 3b and the three-direction cell. The reduced form (Table 7) says the same with the structure controlled.

## Reduced form: bounded and curvature loadings

The RF rows of Table 9 are the `kfh` ladder with bounded (tanh) and curvature (square) loadings on observed characteristics, and the curvature ladder with the loading directions orthogonal to size (`kfo`); the main table (Table 7) keeps only the linear loadings and the latent products.

Bounded loadings behave like linear ones: no room at $K = 1$ and 4, a little at $K = 16$ (12%), and a negative gap throughout, because tanh of a single index is monotone in that index and nearly linear in its rank. Pure curvature of an observed characteristic gives room of about five times the linear ceiling and a gap of about +0.5 at $K = 1$. In the unconstrained square rows the gap then falls to +0.12 and +0.14 at $K = 4$ and 16 while the room does not move: the squared index contains size$^2$, Fama–French's legs are value-weighted and market value is exp(size), so those legs hold a size$^2$ exposure mechanically and Fama–French's Sharpe jumps from 0.05 to 0.46. The `kfo` rows remove the size coordinate from every direction and the gap returns to about +0.5 at every $K$. Curvature of an observed characteristic is therefore captured by random features and by nothing linear, and the square ladder's apparent fall with $K$ was an artefact of value weighting.

The sign ladder (`lsh` rows) shifts the mean of one or both of the two latent factors in the $K = 1$ product by 0, 1, 2 or 3 standard deviations, and `lrt` replaces the product by a one-signed ratio of the two (a debt-service coverage ratio with a ridge). The mean-zero product has room of +0.39 (no noise) and +0.42 (noise 1.0) and fair gaps of +0.031 and +0.093, the $K = 1$ latent-product cell of the reduced-form table; every shifted cell and the ratio have room of 0.000 to 0.005 and a fair gap of −0.005 to −0.014, at the same $SR_{max}$ of 0.598. The linear-in-latents $R^2$ of the loading goes from 0.01 at no shift to 0.79 at a two-sd shift of one factor and 0.89 at two-sd shifts of both. Room in a one-factor economy is therefore all or nothing: a linear portfolio with any nonzero covariance with the exposure mimics the factor, diversifies the rest and takes its whole Sharpe, and only an exposure orthogonal to every linear function of the characteristics leaves the factor to the nonlinear estimator. This is what the structural sign-switching economies of Section 3b test.